The Effect of Unconditional Cash Transfers on the Labor Supply of Workers with Disabilities*

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Abstract

Quantifying income effects is critical for evaluating the efficiency of social insurance and income support programs. However, little is known about the extent to which income effects vary with health status or work impairments. In this paper we provide novel estimates of income effects for workers with permanent partial disabilities. Exploiting a 2005 reform to unconditional workers' compensation benefits in Oregon, we use administrative data on claims and employment to identify income effects by relating differences in benefits to differences in labor supply between observationally identical people before and after the reform. We find that increasing benefits by $1,000 decreases the probability of work by 0.188 percentage points, persisting at least three years after claim closure, and that beneficiaries spend 68 percent of the additional income on leisure. Putting our results in context of a simple utility maximization framework and the broader literature, the findings are large and consistent with income effects being a function of a rapidly increasing disutility of work.

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1. Introduction

Income effects are an important consideration in the optimal design of public tax and transfer programs (Baily 1972, Chetty 2008). While the substitution effect reveals the extent to which changes in the shadow price of leisure distort workers' labor supply, the income effect reflects a non-distortionary response as workers re-optimize their leisure-consumption bundle in response to a change in non-labor income. As a result, income and substitution effects have very different implications for public policy, and quantifying income effects is critical for evaluating the efficiency of social insurance and income support programs, including Social Security Disability Insurance (SSDI), Workers' Compensation and (hypothetically) universal basic income. Several convincing studies provide estimates of income effects for general populations —that is, largely healthy populations—in the U.S. (e.g., Imbens et al. 2001, Golosov et al. 2021) and Europe (e.g., Cesarini et al. 2017, Picchio et al. 2018). But tradeoffs between leisure and consumption depend on individuals' preferences, and specifically on their disutility of work, meaning that income effects likely vary across different populations in important ways. In particular, disutility of work may vary with an individual's underlying health. However, estimating income effects among working-age people with significant health problems has proven challenging, largely due to the strong work disincentives inherent in the programs serving this population.

In this paper, we provide to our knowledge the first estimates of income effects for a population of workers with permanent disabilities unencumbered by programmatic work disincentives. To do so, we take advantage of a 2005 reform to the permanent partial disability (PPD) benefit formula for workers' compensation in Oregon. A key feature of our setting is that PPD benefits are calculated and paid at the end of the workers' compensation claim without any contingencies on future work. Benefits are one-time awards and are either provided in one lump-sum or in payments spread out over three to four months. In other words, the PPD benefit provides a one-time increase in income to the beneficiary without distorting incentives to work, allowing us to identify an income effect.

In 2005, Oregon implemented a major reform that changed the calculation of PPD benefits for all new beneficiaries. Importantly, although all beneficiaries were affected by the change in benefits, the extent to which they were affected varied depending on their characteristics and their injury. We leverage this exogenous change in benefits – as well as variation in the size of this change - to identify the causal effect of an unconditional cash transfer on PPD beneficiaries' subsequent labor supply. Using comprehensive administrative data on workers' compensation claims, disability ratings, and employment records in Oregon, we implement a formula-based instrumental variables approach (Borusyak and Hull 2023). We instrument for benefit values with a rich set of formula inputs measured consistently before and after reform for all claims, interacted with the exogenous policy regime change in 2005.

We find that a $1,000 increase in the PPD benefit amount leads to a 0.188 percentage point (0.26%) decrease in the probability of work. This effect is persistent through at least the first three years after the end of a workers' compensation claim (corresponding to, on average, four years after the onset of disability), well after the majority of any applicants pursuing SSDI benefits would have received a determination (Autor et al., 2015). This suggests a meaningful medium-run labor supply response that is unlikely to reflect strategic behavior while pursuing benefits from other programs. The same $1,000 increase in PPD benefits leads to a reduction in annual hours of 2.36 (0.21%) and a reduction in annual earnings of $38.56 (0.19%), unconditional on employment. A one-time $1,000 increase is equivalent to a $57 per year increase in an annuitized benefit.[1] Thus, we estimate that PPD beneficiaries spend nearly 68% of their additional income on increased leisure time. Considering that we identify a persistent labor supply response to a one-time change in income, these effects are large.

To put these results in context, we employ a simple static utility maximization framework and derive an expression for the income effect showing that that the sensitivity of one's labor supply to non-labor income is a function of the wage (i.e., the shadow price of leisure) and the relative curvature of the utility function with respect to labor supply vs. consumption. Intuitively, the more steeply one's utility falls with labor supplied, the more an individual will find it attractive to use an increase in non-labor income to purchase additional leisure. Our estimate is consistent with a rapidly increasing disutility of work and corroborates prior research findings that workers with permanent partial work-related impairments continue to experience high rates of pain after returning to work (Sears et al. 2021).

Next, we compare our results to those of prior studies focused on estimating the causal effect of non-labor income on labor supply in various populations. Several papers identify income effects in general populations by exploiting unexpected lottery winnings (Imbens et al. 2001, Cesarini et al. 2017, Picchio et al. 2018, Golosov et al. 2021) or unique cash transfer policies (Jacob and Ludwig 2012, Feinberg and Kuehn 2018).[2] Studies from European settings tend to

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[1] We assume a discount factor of 2.4% and an average onset age of 43 to calculate the equivalent amount of an annuity that beneficiaries would receive until the full retirement age of 66. Despite the fact that annuities insure against the risk of outliving one's assets, several studies have documented that individuals tend to prefer lump sum pension settlements to annuities (e.g., Hurd et al, 1998; Brown, 2001; Butler and Teppa, 2007). Thus, our conversion may overestimate the annuity value of the lump sum payment and therefore underestimate the equivalent annuitized income effect.

[2] Jacob and Ludwig (2012) exploit a housing voucher lottery to estimate a negative effect of housing vouchers on labor supply. Feinberg and Kuhn (2018) estimate a negative labor supply response to year-to-year variation in the Alaska Permanent Fund (APF) dividend. Jones and Marinescu (2021) also analyze the effect of the APF on labor supply, but focus on the *macroeconomic* effects of the APF; they find no significant employment effect on the extensive margin and a moderate *increase* in the share of workers in part-time employment, suggesting some reduction in full time work on the intensive margin.

find small or negligible labor supply responses, especially on the extensive margin. On the other hand, studies from American settings tend to find larger income effects, suggesting that differences in population characteristics could matter for the size of income effects. For example, Cesarini et al. (2017) estimate the effect of receiving one million Swedish krona on the probability of employment is -2.015 percentage points; in U.S. dollars, the effect of a $1,000 increase in non-labor income on employment is -0.014 percentage points. Golosov et al. (2021) estimate the same effect at -0.037 percentage points in the U.S.

Focusing on individuals with disabilities, two noteworthy studies by Marie and Vall Castello (2012) and Gelber, Moore and Strand (2017) exploit kinks or nonlinearities in formulae for disability benefits to estimate the effect of the level of monthly disability benefits (received until the full retirement age) on the labor supply of public disability insurance beneficiaries in Spain and the U.S., respectively. Relatedly, Autor et al. (2016) examine the effect of the 2001 Agent Orange decision that expanded eligibility of U.S. veterans with type 2 diabetes who served in the theater for Disability Compensation (DC) benefits on labor supply. Finally, Deuchert and Eugster (2019) exploit a 2004 reform in Switzerland that changed the benefit schedule for DI beneficiaries around age 50 with partial disability ratings between 67 and 69 percent, independently varying their income and the shadow price of leisure, in order to estimate income and substitution effects. If we rescale estimates from these studies to represent the effect of a one-time $1,000 increase in benefits on employment, they range from -0.023 percentage points (Marie and Vall Castello, 2012) to -0.182 (the upper bound estimated by Deuchert and Eugster (2019)). Comparing these estimated income effects to estimates from general populations, it is clear that individuals with disabilities tend to be more sensitive to non-labor income. At the same time, the income effects identified in studies of disabled populations (with the exception of

Deuchert and Eugster) are likely *under*estimates, as the Spanish PPD, SSDI and DC programs all have other parameters that impose work disincentives that may limit beneficiaries' labor supply responses to the specific policy changes used in those studies.[3] Our estimates – coming from a setting where beneficiaries are completely free of any coincident substitution effects – are even larger than this body of work, consistent with the likelihood that prior findings are underestimates.

The insight that income effects depend on the shape of the disutility-of-work function—specifically, the interactions between health and working conditions—may at least partially explain why prior estimates of income effects tend to be larger in American settings that in European settings. It has long been known that Americans tend to have worse health than Europeans, and these health differences are not an artifact of survey measurement (Banks et al. 2006). Recent comparisons between American and European working conditions using harmonized surveys show that American workers tend to work longer hours, face greater cognitive, physical and social job demands, and have greater exposure to posture-related, ambient, and biological/chemical risks (Eurofound and ILO 2019). To test whether income effects are heterogenous by work disutility within our setting, we incorporate an objective measure of the general physicality of one's pre-injury occupation into our analysis using data from the O*NET occupational database. Though suggestive, we find that the income effect increases with this proxy measure of work disutility.

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[3] For example, benefits in the Spanish DI program studied by Marie and Vall Castello (2012) are suspended if beneficiaries start working (Silva and Vall Castello 2012). Similarly, all workers receiving any benefits on either side of the primary insurance amount kink in SSDI examined by Gelber, Moore and Strand (2017) are still subject to the "substantial gainful activity" limit in order to receive benefits leading the authors to note that their estimate reflects a lower bound. Autor et al. (2016) report that 15% of DC beneficiaries are deemed "Individually Unemployable" and concede that for these veterans the estimated labor supply effect likely encompasses both income and substitution effects.

Our paper makes at least three important contributions to the literature. First, we show that income effects are, at least in part, a function of individuals' aversion to additional work. This finding has implications for the broader literature on income effects. For example, it suggests unhealthy populations may benefit more from generous income support programs than healthy populations. Moreover, although existing studies have already made the case that income effects make up most of the total labor supply effect of disability insurance programs (e.g., Gelber, Moore and Strand, 2017), our findings point to a reason why this is likely the case—specifically, that DI beneficiaries' disutility of work is rapidly increasing in work and therefore the benefits are generally well-targeted towards those for whom work is painful. Second, we estimate income effects for individuals with permanent disabilities in the first setting to our knowledge with a *completely* unconditional cash transfer. The fact that our estimate is *higher* than existing (under)estimates among fully disabled populations provides further evidence that disability insurance benefits are largely non-distortionary.

Finally, we focus on an understudied, policy-relevant population—individuals with partial rather than full disabilities—that is likely to grow in the near future as recent American cohorts experience more pain and worse health outcomes than prior cohorts (Institute of Medicine 2007, Case and Deaton 2020, Briggs and Vassall 2021, Brown and O'Brien 2021, NIHR 2022). In our setting, average whole-body impairment ratings range between 5 and 10 percent. PPD beneficiaries are of prime working age, approximately 10 years younger than the typical SSDI beneficiary. Because their impairments are less severe, they may retain a higher work capacity than those deemed permanently totally disabled. Individuals with partial disabilities are also generally excluded from the federal disability insurance in the U.S. At the same time, several policy proposals advocate that partial benefits should be incorporated into future reforms to the disability insurance system in the U.S. (e.g., SSAB 2006, Mitra 2009, Fichtner and Seligman 2016, Maestas 2019). A better understanding of income effects for the population of workers with permanent partial disabilities in particular is thus essential to inform the optimal design of any such future reforms.

The remainder of the paper proceeds as follows. Section 2 provides institutional background on PPD benefits in Oregon and explains how the 2005 reform affected the calculation of PPD benefits. Section 3 describes the data. Section 4 explains our empirical strategy and the necessary assumptions for causal identification. Section 5 presents the main results. In Section 6, we derive an expression for the income effect as a function of the wage and preferences for work and consumption using a simple static utility maximation framework; using this result, we put our estimate into context with estimates from the prior literature and examine heterogeneity in income effects using a proxy measure of work disutility based on claimant's occupation. Section 7 concludes.

2. Institutional Background

In Oregon, when a worker files a workers' compensation claim, she first receives temporary total disability (TTD) benefits equal to two-thirds of wages (subject to a minimum and maximum) after a three-day waiting period from the date of injury to cover missed work time due to the injury.[4] The worker may receive temporary benefits as long as a doctor verifies that she is currently unable to work and her condition has not yet stabilized. Eventually, the worker is deemed to have reached "maximum medical improvement" (MMI), the point where no further

[4] See https://www.oregonlaws.org/ors/656.210 for the exact details of how TTD payments are calculated. Workers' compensation beneficiaries are also immediately eligible for health insurance, which covers any medical expenses associated with the workplace injury.

recovery is expected. At this stage, if there is any residual incapacity due to the injury or illness, the worker is assessed for permanent disability benefits and the claim is closed. On average, it takes just under a year for a claim to resolve, from date of injury to date of claim closure.

The most common type of permanent disability benefit, and the focus in this paper, is the permanent partial disability (PPD) benefit.[5] If awarded, PPD benefits are provided to the worker at the time of claim closure, regardless of the workers' subsequent work activity.[6] Conditional on receiving PPD benefits, the average time from injury date to claim closure date is just over one year, and 95 percent of workers reach MMI within three years.

In Oregon, PPD beneficiaries' injuries are rated along two dimensions: *impairment severity*, or the extent to which the injury results in impairment in functioning, and (if eligible) *work disability*, or the extent to which the injury might prevent future work. The work disability rating takes into account the worker's age, education, the specific vocational preparation required to perform the pre-injury job, and the relationship between the claimant's base functional capacity (before the injury) and residual functional capacity (after the injury).[7]

In 2003, Oregon passed Senate Bill 757 (SB 757), which introduced a significant change to the PPD benefit formula effective for injuries occurring on or after January 1, 2005. This reform changed how the impairment and work disability percentages appear in the benefit formula, but

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[5] In 2018, approximately 17 percent of all indemnity claims in Oregon were for PPD awards. The other main type of permanent benefit is permanent total disability, which accounted for 2 percent of all indemnity claims in 2018. See https://www.oregon.gov/dcbs/reports/compensation/indemnity/Pages/index.aspx.

[6] Awards totaling less than $6,000 are provided in a lump sum at claim closure. By default, larger awards are provided in monthly installments at a rate similar to the temporary benefit rate, although workers with larger awards may opt to receive their PPD benefit as a lump sum. We do not observe whether the beneficiary received a lump sum or monthly payment in our data.

[7] Eligibility for a work disability rating is conditional on the worker being assessed as unable to return to the job held at the time of injury. The exact formula for calculating work disability rating can be found under Rule 436-035-0012 of the Disability Rating Standards Oregon Administrative Rules published by the Oregon Department of Consumer Business Services Workers' Compensation Division.

did *not* change the way these percentages are assessed.[8] Below we explain how PPD benefits were calculated before and after the 2005 reform, and how the reform affected different types of beneficiaries based on their characteristics (specifically, injury type, impairment severity, pre-injury weekly wage, and characteristics associated with work disability eligibility/rating such as age and occupation).

2.1 PPD Benefit Calculation Before the Reform

Prior to 2005, the PPD benefit formula depended on whether the injury involved particular body parts that were listed on a pre-existing schedule ("scheduled injuries") or not ("unscheduled injuries"). For example, injuries to the hand or foot, or hearing loss, were scheduled injuries; unscheduled injuries included conditions such as back pain, shoulder pain, and mental conditions. For scheduled injuries, the extent of impairment was determined relative to the injured body part, whereas for unscheduled injuries, the extent of impairment was determined relative to the whole body. However, both ratings can easily be converted to percentages of the whole body.[9] In addition to being rated for impairment severity, individuals with unscheduled injuries were also potentially eligible for a work disability rating (described above); those with scheduled injuries were ineligible for work disability.

Specifically, the benefit formula prior to the 2005 reform was as follows:

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[8] Note that, before 2005, impairment and work disability ratings were expressed as degrees, which we convert into percentages by dividing by 320 (the maximum number of degrees) and multiplying by 100. After 2005, both ratings were expressed as percentages. For simplicity, we refer to the impairment and work disability ratings as percentages when we describe the benefit formulas before and after the reform.

[9] Rule 436-035-0008 of the Disability Rating Standards Oregon Administrative Rules published by the Oregon Department of Consumer Business Services Workers' Compensation Division explains how impairment ratings for body parts can be converted to impairment ratings for the whole person by dividing the specified degrees for the body part(s) by 320 (the maximum number of degrees for the whole body).

$$PPD_{iT}^{Pre} = (S_i = 1) * \left(BEN_T^S * p_i^p\right) + (S_i = 0) * (BEN_{1T}^U * \min\{p_i^{P+W}, 20\} + BEN_{2T}^U * min\{max\{p_i^{P+W} - 20{,}0\}, 50 - 20\} + BEN_{3T}^U * max\{p_i^{P+W} - 50{,}0\}), \quad (1)$$

where PPD_{iT}^{Pre} denotes the pre-reform PPD benefit awarded to worker *i* for an injury occurring in year *T*, which is a function of whether the injury is scheduled ($S_i = 1$) or unscheduled ($S_i = 0$). Let p_i^P denote the individual's impairment rating expressed as percentage of the person (0-100) and let $p_i^{P+W} = p_i^P + W_i * p_i^W$ denote the sum of the impairment rating p_i^P and, if eligible for work disability, $W_i = 1$, the work disability rating p_i^W.[10] For scheduled impairments, the benefit increased linearly in the impairment rating p_i^P with slope BEN_T^S. For unscheduled impairments, the benefit was a convex kinked function increasing in p_i^{P+W} with kink points at 20 and 50 percent of the whole person. The dollars per percentage (BEN_T^S, BEN_{1T}^U, BEN_{2T}^U, and BEN_{3T}^U) were updated every two years on January 1 in even-numbered years.[11]

2.2 PPD Benefit Calculation After the Reform

In 2005, SB 757 introduced a new rating procedure and benefit calculation to be applied to all PPD cases, eliminating the distinction between scheduled and unscheduled injuries. After the change, all impairments were rated relative to the whole person. Additionally, all claimants, regardless of injury type, are potentially eligible for work disability (if deemed unable to return to the pre-injury job). Finally, if eligible for work disability, the benefit now depends on an

[10] Though technically the work disability rating is a function of several factors including age and occupation, the rating procedure is the same before and after the reform, and we observe p_i^{P+W} in our data consistently throughout the sample period.

[11] In 2004, these amounts were, respectively: $1,788.80 (scheduled) and $588.80, $1,027.20, and $2,393.60 (unscheduled). See Oregon Workers' Compensation Division Bulletin 111 for details of the PPD benefit formula by injury date from before 1992 until the present: https://wcd.oregon.gov/Bulletins/bul_111.pdf (accessed 10/2/23). We convert dollars per degree to dollars per percentage by scaling by 320/100.

additional factor: the individual's pre-injury weekly wage. Specifically, the benefit formula after the 2005 reform is as follows:

$$PPD_{iT}^{Post} = p_i^P * SAWW_T + W_i * p_i^{P+W} * 1.5 * w_{iT}. \qquad (2)$$

where PPD_{iT}^{Post} indicates the benefit post-reform, $SAWW_T$ is the state average weekly wage in the year of injury, and w_{iT} is the individual's pre-injury weekly wage (subject to a minimum of 50% and maximum of 133% of $SAWW_T$).[12]

In the next subsection, we describe how the reform affected PPD benefits based on individual characteristics such as injury type, impairment severity, and pre-injury weekly wage. In the following section, we explain how we leverage this variation in the effect of the reform on PPD benefits by individual characteristics to identify the causal effect of PPD benefits on labor supply.

2.3 Variation in the Effect of the Reform on PPD Benefits by Individual Characteristics

Table 1 illustrates the effect of the 2005 reform on PPD benefits for select cases. For each case, we assume the beneficiary has a pre-injury weekly wage of $600. In the first case, we assume that an individual receives an injury to their arm (a scheduled injury) resulting in an impairment rating of 12% of the person; furthermore, we assume that the individual is not eligible for work disability after 2005. Applying the scheduled benefit formula for 2004, the individual would receive $21,466 in PPD benefits if the injury occurred in 2004; however, this same individual would receive only $8,263 if the injury occurred in 2005, *with no other differences in individual or injury characteristics*.

[12] On January 1, 2005, the state average weekly wage was $688.56 and was updated every July 1 thereafter.

The second case changes the location of the injury, from the arm (scheduled injury) to the back (unscheduled injury), leaving all other features (impairment rating, work disability rating, etc.) the same as in the first case. Applying the 2004 benefit formula for an unscheduled injury, this individual would receive $7,066 in PPD benefits; because the 2005 reform erased the difference between scheduled and unscheduled injuries, the individual would receive $8,263 in PPD benefits if the injury occurred in 2005—the same benefit amount as an individual with a scheduled injury (case 1).

The next two cases illustrate the effect of eligibility for a work disability award on PPD benefits before and after the reform, for scheduled vs. unscheduled injuries. In the third case, we again consider an individual with an arm (scheduled) injury with an impairment rating of 12% but now we assume they would be eligible for a work disability award after 2005 (recall that before 2005, individuals with scheduled injuries were ineligible for work disability). Before 2005, this individual would receive a PPD award of $21,466, the same as someone without a work disability rating (case 1); after 2005, however, the individual would receive $24,463 in PPD benefits—a $2,997 increase. For an individual with a back (unscheduled) injury who is otherwise identical to the individual in case 3, benefits would be $10,598 if they were injured in 2004, but $24,463 if injured in 2005 (case 4).

Finally, the last two rows of Table 1 illustrate how the balance of impairment vs. work disability rating affects the difference in PPD benefits before and after the reform for individuals with unscheduled injuries. Note that the two individuals in cases 4 and 5 have the same PPD benefit level before 2005 because their impairment and work disability ratings both add up to the same number, 18%. However, after 2005, the two ratings enter the PPD benefit formula separately (rather than as a sum). As a result, the two individuals receive different benefit amounts after the reform—in this case, the individual with the lower impairment rating and higher work disability rating (case 4) receives $4,132 more than the individual with the higher impairment rating and lower work disability rating (case 5) after 2005 (i.e., $24,463-$20,331).

Figure 1 illustrates the effect of the 2005 reform on PPD benefits more generally. We use the benefit formula to calculate the benefit before 2005 (solid line) and after 2005 (dashed line) for each possible impairment rating, separately for injuries which, based on the body part, would have been classified as scheduled (left column) and unscheduled (right column) prior to the reform. Because benefits vary with multiple parameters we plot three cases: 1) no work disability rating; 2) a 10% work disability rating and pre-injury wage of $400 (the 25th percentile in our sample); and 3) a 10% work disability rating and pre-injury wage of $800 (the 75th percentile in our sample). For each case-type, the difference between the solid and dashed lines represents the hypothetical difference in benefits for workers injured before and after the policy change, while holding all characteristics fixed.[13] Both the sign and magnitude of the hypothetical change in benefit due to the reform vary enormously, ranging from -$75,000 (in very rare cases) to upwards of $45,000 depending on the combination of the type of injury, impairment rating, potential work disability rating and the claimant's pre-injury wage. Prior to the reform, individuals with scheduled injuries tended to receive higher PPD benefits than those with unscheduled injuries, especially if the unscheduled injury was not rated for work disability; however, the reform equalized benefits for those with scheduled and unscheduled injuries, with the presence of work disability (newly available to those with scheduled injuries), along with pre-injury wage, now being the primary driver of differences in benefit levels.

[13] Figure A1 plots the differences corresponding to the benefit levels shown in Figure 1.

3. Data Sources

Detailed data on injury types, disability ratings, and worker characteristics are essential to examine the effect of this policy change. We use several administrative datasets from the Workers' Compensation Division of the Oregon Department of Consumer and Business Services (DCBS) and the Oregon Employment Department (OED). DCBS provided claim-level data for all closed claims for workers' compensation indemnity benefits between 1987 and 2012. The database includes information about total indemnity and medical payments made on the claim, and key dates including date of injury, first and last dates of total temporary disability (TTD) payments, and claim closure date. Worker characteristics included in the database are date of birth, gender, pre-injury weekly wage, industry and occupation. DCBS also provided information on total permanent partial disability (PPD) awards, injured body parts, and award type (i.e., impairment, work disability). Additional information about return to work at the time of claim closure is provided for the subset of claims with injury years between 2001 and 2012. The dataset also includes impairment ratings for injury years 1999-2012.

DCBS worked with OED to match PPD awards in these years to quarterly wage records in the state Unemployment Insurance (UI) database starting in the third quarter of 1999. We obtained employment data through the fourth quarter of 2013. DCBS and OED matched records using worker Social Security Numbers and excluded outlier records in the wage database as well as observations with inconsistent and incomplete data. OED and DCBS achieved a 97 percent worker match rate between the UI database and workers' compensation claims records. The UI database includes quarterly data on total earnings, hours and an anonymized employer ID.

Together, these data sources give us a detailed account of claimant demographic and injury characteristics, PPD rating and other formula inputs used to calculate PPD benefits before and after the reform. Additionally, we have complete employment information before and after injury for closed PPD claims in Oregon between 2001 and 2013. Because PPD claims can take years to develop and ultimately close, we apply a constant maturity screen to all injury years in our analysis and include claims that were closed within two years of the date of injury. This screen addresses concerns that slow-developing claims in later injury years might not have closed at the time of the match to the wage records and would be disproportionately excluded from the dataset. We restrict our analysis sample to injuries occurring between 2001 and 2009 (that closed by the end of 2010) to observe post-claim labor supply for up to three years after the claim closes (by 2013, the last year for which we have employment data). In our sample, claims last on average approximately one year, meaning we observe labor supply outcomes up to four years after the onset of disability. After these restrictions, our total sample size is approximately 34,000.[14]

Column 1 of Table 2 presents descriptive statistics for all observations in our sample. Sixty three percent of claimants are older than age 40 and 72 percent are men. The average pre-injury weekly wage is $625 in 2005 dollars ($967 in 2023 dollars). Total medical expenditures averaged $12,677 in 2005 dollars ($19,649 in 2023 dollars). Temporary total disability (TTD) benefits were paid for 50 days before claim closure, on average. Slightly more than 80 percent of claimants in the database were released to work by a doctor at the time of claim closure, and just over two-thirds had returned to work prior to claim closure. Over half of claims occur in one of four main occupation categories: transportation (18%), production (17%), construction (12%), and maintenance (7%). Approximately one-third of all claims result from fractures and breaks,

[14] Applying a constant maturity screen of three years (restricting the sample size to 38,000) allows us to observe labor supply up to two years after claim closure and yields similar results (available upon request).

and one-third from muscle strains or sprains. Trauma and unexpected injuries account for 15 percent of cases, followed by wounds, cuts and burns (8 percent), with other injuries making up 9 percent of claims.

4. Empirical Strategy

4.1. Instrumental Variables Research Design

The goal of our paper is to estimate the causal effect of PPD benefits on labor supply outcomes of beneficiaries. From a glance at Figure 1, a natural estimation strategy appears. Observationally identical individuals—those with the exact same injury, demographic characteristics and pre-injury job—received in some cases substantially different PPD benefit amounts based solely on whether the individual was injured before or after January 1, 2005. As a result, we would expect to see a much larger effect of the reform on the labor supply for worker-types with larger differences in benefits before and after the reform.[15]

Formally, in order to exploit the wide variation in the effect of the 2005 reform on PPD benefit levels, we implement a formula-based instrumental variables approach, similar to the approach described in Borusyak and Hull (2023). Specifically, we estimate a two-stage model of the following form:

$$E[PPD_{iT}] = POST_{iT} * Z_i * \psi + Z_i\,\gamma + \lambda_T. \tag{3}$$

$$E[y_{it}] = \phi PPD_{iT} + Z_i\,\beta + \delta_T. \tag{4}$$

In the first stage, we predict the benefit PPD_{iT} for worker i who was injured in year T using information about a comprehensive set of observable demographic and case characteristics Z_i

[15] Note that the individuals themselves do not directly experience income gains or losses, so we would not expect there to be asymmetric responses for gains vs. losses.

(described in detail below), interactions between Z_i and an indicator for claims that occurred after 2005 ($POST_{iT}$* Z_i), and injury year fixed effects λ_T.[16] Note the instrumental variables are the set of *interactions* between worker/injury characteristics and post-reform indicators ($POST_{iT}$* Z_i). We use the benefit formulas shown in equations (1) and (2) to determine which variables and interactions belong in equation (3), as well as the appropriate functional forms (e.g., splines in the sum of impairment rating and work disability rating for unscheduled injuries to account for the convex kinked benefit function prior to the reform). In the second stage, we regress labor supply outcome y_{it} in post-closure year t (i.e., t years after the claim was closed) on predicted benefits from equation (3), along with controls for observable characteristics and injury year fixed effects. The coefficient ϕ represents the causal effect of an increase in PPD benefits (unconditional income) on labor supply.[17]

To illustrate how our approach identifies the causal effect of PPD benefits on labor supply, consider a case with only two types of people, classified by a single binary indicator variable Z_i, and two periods, where $T=0$ denotes the pre-reform period and $T=1$ the post-reform period. Normalize $\lambda_0=0$. From equation (3), we see the expected "dose," or *difference* in benefits for two *individuals who only differ in their injury date and are otherwise identical*, is λ_1 if $Z_i=0$, and $\psi+\lambda_1$ if $Z_i=1$.[18] From equation (4), we see the expected "response," or difference in labor supply, is $\phi\lambda_1+\delta_1$ if $Z_i=0$, and $\phi(\psi+\lambda_1)+\delta_1$ if $Z_i=1$. Taking differences across the two types of

[16] More precisely, we condition on and interact observable characteristics with a *series* of "policy regime" fixed effects based on injury year to account for different benefit schedules based on changing factors within regime (i.e., scheduled and unscheduled degrees per dollar, state average weekly wage).
[17] Similar approaches have been used to study the regional effects of the federal minimum wage (Card 1992), the impact of student aid on college enrollment (Nielsen et al. 2010), and the effect of disability insurance benefit generosity on labor supply in Austria (Mullen and Staubli 2016).
[18] If $Z_i=0$, expected pre-reform benefits are λ_1 and post-reform benefits are 0 (since we normalize $\lambda_0=0$); the difference is therefore λ_1. Similarly, if $Z_i=1$, expected pre-reform benefits are $\gamma+\lambda_1$ and post-reform benefits are $\psi+\gamma$; the difference is $\psi+\lambda_1$.

individuals, the difference in dose is ψ and the difference in response is $\phi\psi$. The ratio of the differences is ϕ.[19]

To implement this approach, we need consistent measures of formula inputs for all workers before and after the policy change. However, a limitation of administrative claims data, both generally and in our setting, is that, while it includes rich information on demographic and case characteristics, it only captures the specific formula inputs—or combination of inputs—required to calculate benefits in the contemporaneous policy regime and not in alternative regimes. In particular, prior to 2005, work disability ratings (if any) were not reported separately from impairment ratings for unscheduled claims since only the sum of the ratings was needed to calculate benefits. Moreover, since scheduled claims were ineligible for work disability awards prior to 2005, we do not know which of these claims would have been eligible for work disability, or what the rating would have been. Finally, due to the change in focus from ratings based on individual body parts to those based on the whole person, there are some differences in the way specific body parts are recorded in the database before and after the reform.[20]

We account for these asymmetries by using in our analysis only those observable variables that are available and consistently recorded for *all* claims in the data set—both pre- and post-

[19] As pointed out by Callaway et al. (2024), if ϕ is heterogeneous, then identification based on a dose-response relationship implicitly assumes the "dose" is randomly assigned with respect to the potential treatment effect (ϕ). In our setting, since the dose ($POST_{iT} * Z_i * \psi + \lambda_T$) is determined by a complex combination of individual- and regime-specific factors (i.e., impairment type, impairment rating, eligibility for work disability, work disability rating, and pre-injury wage as well as scheduled and unscheduled degrees per dollar and state average weekly wage), the treatment effect is unlikely to systematically vary with the dose. See also Appendix C.1 of Borusyak and Hull's 2021 working paper, which discusses the conditions under which a formula-based IV strategy has a casual interpretation with continuous treatment (in our case, benefits).

[20] For example, suppose a worker burned her hand. Prior to the reform, each finger would receive a separate scheduled rating based on the extent of the burn to that finger, and the multiple injuries would be combined to obtain a total scheduled award. After the shift to assessment of impairment for the whole person, the distinction between hand and finger no longer mattered, so the same burn would more likely be reported simply as an injury to the hand.

2005. Specifically, we include in Z_i the impairment rating interacted with an indicator for scheduled injuries and the *sum* of the impairment and work disability ratings interacted with an indicator for unscheduled injuries. To address the issue of missing (i.e., non-separate) work disability eligibility and rating in the pre-reform period, we include in Z_i a comprehensive set of demographic and case characteristics available in our data that predict work disability eligibility and rating, including as whether the claimant returned or was released to work by a physician prior to claim closure, age, gender, injury type, medical expenditures, TTD duration and occupation categories.[21] We use this set of characteristics instead of the actual work disability rating even in cases where we observe the work disability rating to ensure consistency in our prediction across all claims. Finally, to account for differences in injury records across regimes, we aggregate individual injuries into broader body systems (e.g., combining injuries to the hand and fingers). We derive the specification for the first stage equation (3) from the known PPD benefit schedules in equations (1) and (2), replacing inconsistently observed variables with their consistently observed predictors.[22] The first stage regression has an F-statistic of 233.5, indicating that the instruments collectively are strongly predictive of the actual benefit difference (i.e., we reject the hypothesis that the coefficients ψ on $POST_i * Z_i$ are zero at very low significance levels).

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[21] As discussed in Section 2, workers are eligible for work disability if they are deemed unable to return to their pre-injury job. Conditional on eligibility, the work disability rating is a function of the worker's age, education, the specific vocational preparation required to perform the pre-injury job, and the relationship between the claimant's base functional capacity (before the injury) and residual functional capacity (after the injury). Our data set does not include education or claimants' base and residual functional capacity, but it does include a rich set of demographic, injury and occupation characteristics that are correlated with work disability eligibility and rating observed in post-2005 claims (see Tables A1 and A2).

[22] Note that if we could perfectly predict benefits in both regimes then the first stage is perfectly identified and two-stage least squares will break down; in that case, a control function strategy can be employed to identify the causal effect of benefits on labor supply (see e.g., Mullen and Staubli, 2016).

4.2. Assessing Potential Threats to Identification

The above evidence demonstrates that the policy change is associated with large differences in benefit values for observationally identical individuals. In order to identify causal effects with this approach, we also require an assumption of monotonicity and an exclusion restriction. We begin with a discussion of the monotonicity condition and then provide empirical evidence in support of the exclusion restriction.

In our setting, the monotonicity condition requires that conditional on Z_i, the difference in pre- and post-reform benefits is either positive for everyone or negative for everyone. If we could observe all benefit inputs needed consistently pre- and post-reform, then the monotonicity condition would be trivial (i.e., the benefit difference is point-identified).[23] Instead, we must use consistently observed predictors of benefits to form our comparison groups, inducing some measurement error in the first stage; in this case, monotonicity holds if the measurement error in benefits is small enough (i.e., the groups constructed conditional on Z_i include individuals with similar enough benefit differences that they are all positive or negative). Figure 2 plots the observed PPD benefit against the predicted benefit estimated from equation (3), separately for claims before 2005 (Panel a) and after 2005 (Panel b). The predicted benefit lines up exactly with the actual benefit for claims before 2005, as indicated by the fact that all data points fall on the 45 degree line in the chart; this is because all formula inputs required to calculate the pre-2005 benefit are observed for all claims. There is more noise in the predicted benefit for post-2005 claims due to the fact that our controls do not perfectly predict work disability, but the

[23] Figure A2 shows that when we use all available formula inputs in the post-2005 regime only, including the actual work disability rating, we predict actual post-2005 benefits with 100% accuracy.

trend still tracks the 45 degree line closely. Overall, the first stage regression has an R^2 of 0.94, indicating minimal measurement error in predicted benefits.

The other key identifying assumption is that, conditional on Z_i, variation in observed benefits (driven by the policy change, $POST_{iT}* Z_i$) is uncorrelated with unobserved determinants of labor supply (e.g., pre-injury labor force attachment or injury severity). Practically, this assumption may be violated if there are systematic shifts in unobserved claim characteristics before and after the reform, whether by chance or due to benefit-maximizing manipulation by beneficiaries or raters. Note that shifts in characteristics across regimes are only problematic if they are correlated with unobserved labor supply determinants. We conduct several tests to rule out accidental or strategic manipulation, along both the intensive (claim characteristics, injury rating) and extensive (claiming decisions and timing) margins.

First, we investigate whether observable characteristics are balanced before and after the reform. Table 2 compares observable characteristics of claims in our sample with injury dates before and after 2005. We calculate the difference between post- and pre-2005 claims and present p-values from tests of statistical significance for these differences. While several of the differences are statistically significantly different from zero, most differences are small in magnitude and the directions do not point to systematic differences in unobserved labor determinants. Workers after 2005 are more likely to be older than 40 (62 vs. 64 percent), are less likely to be men (73 vs. 71 percent), received TTD benefits for more time (51 vs. 49 days), are more likely to have been released to work prior to claim closure (81 vs. 79 percent), and are more likely to have returned to work prior to claim closure (68 vs. 67 percent). There are also differences in occupations, perhaps reflecting broader macro changes in the economy over the decade. The share of injuries categorized as strains and sprains is 11 percentage points less after the reform, while injuries from trauma are 8 percentage points higher. Furthermore, Figure 3 reveals that most of these differences reflect broader trends in characteristics over time rather than discrete breaks in levels coinciding with the timing of the reform. The figure presents bin-scatter plots by injury month for selected claim characteristics. As shown in the first three panels, the trends in the worker age at injury, pre-injury weekly wage, and share of claims that had returned to work at the time of claim closure are changing smoothly over time, with no evident trend break around the time of the reform.

While we can control for pre- and post-reform differences in observable claim characteristics in our estimation strategy, what matters most for identification is that differences in claim characteristics do not translate into systematic differences in ratings. Table 3 compares the distribution of broad body system injuries and the associated PPD ratings before and after 2005, separately for scheduled and unscheduled injuries to account for the comparability issues discussed in Section 4.1.[24] As shown in Panel A, the average overall impairment rating among scheduled injuries is balanced before and after the reform, and the differences in ratings conditional on injury type are all less than 1 percentage point and not statistically significant (with the exception of a statistically significant 0.2 percentage point difference in ratings for leg/hip injuries). Panel B shows there are no statistically significant differences in combined impairment and work disability rating conditional on injury type among unscheduled injuries. The consistency of the impairment and work disability percentages before and after the reform is reassuring in that, although there appear to be some shifts in the composition of injuries over time, there is no evidence of systematic shifts in ratings coinciding with the timing of the reform

[24] Specifically, for scheduled injuries we show the average impairment rating as a percentage of the person before and after the reform, and for unscheduled injuries we show the average combined impairment and work disability ratings.

that could be indicative of either changes in rater behavior or differences in the unobserved severity of injuries before and after the reform. The last three panels of Figure 3 show that the share of claims with injuries that would be classified as scheduled, as well as the average impairment and work disability impairment ratings, trend smoothly through 2005.

To understand whether the idiosyncratic differences in observable characteristics indicate systematic shifts in unobservable characteristics, it is useful to examine trends in predicted benefits, which provide a summary measure of claim characteristics. Figure 4 plots the trend in hypothetical benefits for each claimant using their observed characteristics holding the benefit formula fixed in 2004 and 2005, respectively. Specifically, in panel (a) we predict the hypothetical PPD benefit from Equation (3) assigning all claimants a hypothetical claim year of 2004 to generate a hypothetical pre-reform benefit for all claimants. We repeat the exercise in panel (b) instead assigning all claimants a claim year of 2005 to plot a hypothetical post-reform benefit for all claimants. Because the formula is held constant in these comparisons, any discrete breaks in expected benefits around 2005 in panel (a) or (b) would result from systematic shifts in the underlying formula inputs. However, the figures again show no evidence of systematic shifts in benefit inputs coinciding with the reform.

Next we address potential manipulation in claiming decisions and timing. First, we first note that the policy regime is determined by the injury date and it can often take a year or longer for workers' compensation claims to reach the point of maximum medical improvement, when claims are rated for PPD. In the early stages of a claim, workers are unlikely to be able to anticipate the extent of eventual permanent impairment or work restriction. Furthermore, few workers are familiar with the details of the program before experiencing an injury and even fewer were likely aware of the details of SB757 to the extent they could anticipate the implications for their own benefits (Rennane and Cherney 2019). As a result, it is unlikely that workers could strategically manipulate the timing of their injury in order to qualify for a more generous benefit. Overall award rates, claim durations, and the share of claims that ultimately received PPD benefits are all similar before and after 2005, although the frequency of claims and the share of claims receiving PPD trend downward trend during the 2000s (DCBS 2015).

Even if claimants do not strategically delay or expedite their claims around 2005, it is still possible that claimants with higher expected benefits could be more likely to file a claim at a given point in time. In this case, we might expect to observe bunching of more claims from those who could expect a larger benefit after the reform ("winners") and fewer claims from those who could expect a smaller benefit after the reform ("losers") after 2005. Because we only observe a claimant's actual benefit and not their benefit in the alternative regime, we test this empirically by using equation again using the predicted hypothetical benefits for all claims before and after the reform as described in Figure 4. Here, we divide the sample into expected winners and losers based on whether the difference between the expected benefit in 2005 and 2004 was positive or negative. Figure 5 presents bin-scatter plots by injury month of the number of claimants, separately for expected winners and losers. In both cases, we find no evidence of a discontinuous change or bunching in claiming behavior before vs. after the reform, and if anything we find there were more claims from expected losers and fewer claims from expected winners after the reform.[25] We formally test whether there was a discontinuous increase in the number of claimants from the winning group and/or decrease in the number of claimants from the losing group after the reform, by regressing claim counts on a post-2005 indicator variable, a flexible polynomial in month of injury and an interaction between the post-2005 indicator and

[25] Figure A3 shows bin-scatter plots for total claim volume over time and we also do not see bunching.

polynomial in month of injury. The coefficients on the post-2005 indicators were not statistically significant, providing no evidence of bunching.

Taken together, we find some evidence of small, idiosyncratic differences in the observable characteristics of claims before and after the reform, but no evidence of changes in the frequency of claims or ratings which might reflect systematic differences or strategic manipulation either by potential claimants or examiners. We conclude that, conditional on observed characteristics, claims are comparable to one another in terms of unobserved severity and other characteristics likely to affect labor supply. As a result, we attribute any observed variation in labor supply (conditional on observable characteristics) to variation in PPD benefit levels resulting from the 2005 reform.

5. Main Results

Figure 6 presents an event study of the effect of PPD benefits on labor supply over time, examining the estimated effect during the eight quarters before injury, the quarter of claim closure, and the twelve quarters after closure.[26] Panel (a) shows the effect of PPD benefits on (unconditional) weekly earnings, panel (b) shows the effect on (unconditional) weekly hours worked, and panel (c) shows the effect on the probability of employment, for each quarter. Each point estimate on the graphs represents the coefficient on the PPD benefit, scaled by $1,000, from equation (4). The eight quarters prior to injury serve as a placebo test since, conditional on case characteristics, the benefit should not be correlated with labor supply before the worker is

[26] We omit the quarters occurring between the injury and claim closure (when beneficiaries are not working).

injured. The effect is very close to zero and not statistically significant in any quarter prior to injury.[27]

We interpret the estimates in the quarters after claim closure as estimates of the causal effects of PPD benefits on labor supply over time. For all three outcomes, the point estimate is largest in the second quarter after claim closure, and the point estimates decline slightly in magnitude over the subsequent ten quarters. In general, however, the effect is quite stable – similar in magnitude and statistical significance for each of the three labor supply outcomes. Because claims close on average approximately one year after injury, these results reflect persistent reductions in labor supply nearly four years after the onset of disability.

Table 4 shows the IV coefficients from models estimating the impact of PPD benefits on labor supply outcomes in the third year after claim closure. Column 1 shows the effect of PPD benefits on the probability of employment, and columns 2 and 3 show the effect of PPD benefits on (unconditional) hours worked and earnings, respectively, during the third year after closure. Across all outcomes, increasing the value of the PPD benefit has a statistically significant negative effect on labor supply, indicating the presence of an income effect. Column 1 shows that increasing the PPD benefit by $1,000 reduces the probability of returning to work by 0.188 percentage points. Columns 2 and 3 of Table 3 show that a $1,000 increase in the PPD benefit results in a reduction in hours worked by approximately 2.4 hours per year and a reduction in annual earnings of $39.[28]

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[27] The quarter of claim closure can be viewed as a "partially treated" quarter, since claims are closed at varying points during the quarter.

[28] Table A3 presents results for log hours and earnings, conditional on employment. We find a small and statistically insignificant labor supply response conditional on employment, suggesting the labor supply response is driven by changes in the extensive margin. This could be because workers do not have much leeway in setting their work hours or schedule and as a result do not have the ability to adjust their labor supply along the intensive margin. Our results are also consistent with a reservation wage model of labor

These results show that the receipt of a sizeable, unconditional cash payment reduces labor supply, providing evidence of an income effect. While the absolute magnitude of these point estimates is small, these estimates reflect a persistent (medium-run) change in extensive margin labor supply in response to a relatively small change in income. As a result, these point estimates reflect a large response to a relatively small benefit. Indeed, assuming a discount factor of 2.4% and that the average PPD beneficiary will work for an additional 23 (=66-43) years, we calculate that a one-time $1,000 increase is equivalent to a $57 per year increase in a hypothetical annuitized benefit. Since a $1,000 increase in benefits decreases average annual earnings by $38.56, we estimate that PPD beneficiaries spend approximately two-thirds of the additional income arising from the 2005 reform on increased leisure time.

6. Income Effects and Work Disutility

To put our results in context, consider the following simple static utility maximization problem:

$$\text{Max } u(c, l)$$

$$s.t. \;\; c = y + wl,$$

where c is consumption, l is labor supply, y is non-labor income, and w is the wage. Assume diminishing marginal utility of consumption and increasing marginal disutility of work ($u_c > 0$, $u_{cc} < 0$, $u_l < 0$, $u_{ll} > 0$), and for simplicity assume utility is separable in consumption and

supply where workers' reservation wages increase in response to additional income, leading some to drop out of the labor market.

labor supply ($u_{cl} = 0$). Differentiating the first order condition with respect to *y*, substituting in $w = -u_l/u_c$, and rearranging terms, we obtain the following expression for the income effect:

$$\frac{\partial l}{\partial y} = \frac{-1}{w\left(1 - \frac{1}{w}\frac{u_{ll}/u_l}{u_{cc}/u_c}\right)}.$$

From the expression, we can see that the sensitivity of one's labor supply to non-labor income is a function of the wage and the relative *curvature* of the utility function with respect to labor supply vs. consumption. Note that if disutility of work is linear in hours worked (i.e., the marginal disutility of labor is constant), then $u_{ll} = 0$ and $\frac{\partial l}{\partial y} = -\frac{1}{w}$; in that case the income effect varies inversely only with the wage (the shadow price of leisure). However, if the marginal disutility of work is increasing in hours worked, then reducing one's labor supply in response to an exogenous change in income becomes even more attractive. Intuitively, the more steeply one's utility falls with labor supplied, the more an individual will find it attractive to use an increase in non-labor income to purchase additional leisure.[29]

The insight that income effects depend on the shape of the disutility-of-work function may at least partially explain why estimates of income effects in the literature vary by order of magnitude.[30,31] Table 5 presents the settings and income effect estimates from nine prior studies, along with the main estimates from this paper, rescaled to represent the effect of a one-time $1,000 increase in non-labor income on the probability of employment, and ordered from

[29] If workers are credit constrained, receipt of non-labor income could also have significant impacts on labor supply by relaxing the constraint in addition to the impacts through the disutility channel. Unfortunately our data do not contain information on savings which would enable us to test this hypothesis, and we are underpowered to use the change in default from lump sum at $6,000 as an alternative test.

[30] We are not the first to note the relationship between disutility of work and the magnitude of the income effect. See, for example, Chetty (2006).

[31] Other factors that could affect the magnitude of the income effect and likely vary across settings include risk aversion, wage rates, and labor demand effects arising from wage subsidies for disabled or other types of workers (Aizawa et al., 2025).

smallest to largest in magnitude.[32] For studies estimating income effects in lottery settings, we simply rescale the published estimates of employment effects to represent a the effect of a $1,000 increase.[33] For studies estimating income effects in social insurance settings (e.g., cases where benefits are received as an annuity rather than one-time), we calculate the average duration of benefit receipt by subtracting the average age of initial benefit receipt from the full retirement age in that setting and, assuming a 2.4% discount factor, convert changes in annuitized benefits to their equivalent of a one-time $1,000 increase in non-labor income.[34] Finally, for the housing voucher study in Jacob and Ludwig (2012), we take the annual cash value of the voucher reported in the paper and assume that recipients consume the subsidy until age 66.[35]

Table 5 reveals two interesting patterns. First, the three studies with the smallest income effects are all studies from European settings. This may reflect both Europeans' better average health and working conditions than their American counterparts (Banks et al. 2006, Eurofound and ILO 2019). Second, income effects estimated in general populations tend to be smaller than those estimated in older and disabled populations, even though prior estimates from studies of

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[32] We exclude studies that do not report employment effects (but instead focus on hour or earnings). We also exclude a recent paper by Artmann et al. (2023) examining the labor supply effects of a 2014 German reform that increased *future* pension wealth for mothers by on average €3830 (4.4%) per child born before 1992; they find a statistically significant and sizable effect of the reform on earnings but not employment, i.e., operating solely through the intensive margin channel.

[33] E.g., for Cesarini et al. (2017), we take -2.015 from Table 4, use the conversion 1 million SEK = $110,000 from the paper, and divide -2.015/110=-0.018.

[34] E.g., Gelber, Moore and Strand (2017) report that their preferred estimate of the effect of a $1,000 increase in DI benefits on employment is -1.3 percentage points (Table 3). Assuming this increase applies to benefits for 16 years (=66-50, the average age from Table 1), this translates to a one-time increase of $13,157. The scaled effect is therefore -1.3/13.157=-0.099. Giupponi (2019) reports that the effect of a €1,000 increase in annual survivors' benefits on labor force participation is -4.4 percentage points (Table 6, column 1, row 1), however she also reports that the change in benefits due to the reform is equivalent to approximately €100,000 in lifetime benefits. We use a Euro-to-Dollar conversion of 1.3 given time frame in the paper, and obtain a scaled effect of -4.4/13000 = -0.034.

[35] Jacob and Ludwig (2012) report the annual cash value of the voucher as $8,160-$3,735=$4,425 (bottom of pg. 281), the average age of the head of household as 32 years (Table 1) and the IV estimate of the effect of the housing voucher on employment of household heads as -0.036 (Table 3).

disability insurance beneficiaries are likely underestimates given the institutional work disincentives present in these settings. The estimate from our paper, from a setting without such disincentives, is the largest estimate in Table 5. Taken together, the estimates from the literature are generally consistent with a hypothesis that income effects reflect differences in underlying disutility of work across populations.

To investigate this relationship further within our setting, we examine heterogeneity in income effects using a proxy measure of work disutility obtained from the Occupational Information Network (O*NET) database published by the U.S. Bureau of Labor Statistics.[36] The O*NET database includes detailed work characteristics for over 900 occupation profiles covering jobs across the U.S. economy. To proxy for work disutility, we use a measure from the O*NET Work Activities module of the level of complexity of performing general physical activities needed to perform one's job, where general physical activities include "activities that require considerable use of your arms and legs and moving your whole body, such as climbing, lifting, balancing, walking, stooping, and handling materials." The measure is collected on a scale from zero to seven, with zero reflecting occupations where physical activities are not at all important and seven indicating jobs with extremely high levels of physical activity. The claims data include two-digit (pre-injury) occupation consistently over our analysis period, so we merge the O*NET physicality measure to our claims at the two-digit occupation level.

Using these data, we interact our measure of the predicted benefit with indicators for the quartiles of physicality to examine how the income effect varies with the disutility of work.[37] Table 6 shows that as the physicality (work disutility) of occupations increases, the income effect

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[36] Since injury severity and work disability ratings are inputs into the PPD benefit formula, we do not investigate heterogeneity by these measures.

[37] In our sample, physicality ranges from 0.93 to 4.8, with 25th, 50th, and 75th percentiles of 3.09, 3.88 & 4.37.

also increases. For workers in the lowest quartile of physicality, the point estimates are smaller and not statistically significant. For higher quartiles, the point estimates are larger than our main results and all statistically significant. While p-values from an F-test cannot reject the equivalence of these coefficients, the pattern is nevertheless consistent with a larger income effect as disutility of work increases.

7. Conclusion

Despite their importance for understanding individual behavior and designing public policy, income effects have been difficult to identify empirically, particularly for populations with disabilities. This paper provides to our knowledge the first estimates of income effects for workers with permanent partial disabilities who do not face institutional work disincentives. We take advantage of a 2005 reform to the permanent partial disability (PPD) benefits formula for workers' compensation in Oregon to estimate the causal effect of non-labor income on labor supply outcomes. We identify the income effect based on the dose-response relationship between differences in benefits and labor supply between observationally identical people whose injuries occurred before and after the reform. Using comprehensive administrative data on workers' compensation claims, disability ratings, and employment records in Oregon, we implement a formula-based instrumental variables approach where we instrument for PPD benefits with a rich set of formula inputs measured consistently before and after reform for all claims, interacted with indicators for the exogenous change in the policy regime.

This analysis yields large and persistent income effects for this population. We find that a $1,000 increase in the PPD benefit amount leads to a 0.19 percentage point (0.26%) decrease in the probability of work. This effect is persistent through at least the first three years after the end of a workers' compensation claim (on average, four years after injury onset), suggesting a meaningful medium-run labor supply response. The same $1,000 increase in PPD benefits leads to a reduction in annual hours of 2.36 (0.21%) and a reduction in annual earnings of $38.56 (0.19%). Considering the fact that we identify a persistent labor supply response to a one-time change in income, these effects are large. We estimate that PPD beneficiaries spend two-thirds of the value of their additional PPD benefits on increased leisure time.

To put our results in context, we derive an expression for the income effect as a function of the wage and preferences for work and consumption using a simple static utility maximation framework. Using this framework, we compare our results with other estimates of income effects in the prior literature for a variety of populations in the U.S. and Europe, and for healthy and disabled populations. We find that our estimate is significantly larger than estimates for healthy populations, and of a similar magnitude to estimates of disabled populations in the U.S. Still, our estimate is slightly larger than other estimates of disabled populations, likely resulting from the absence of broader work disincentives inherent in other disability programs. Furthermore, we find evidence that heterogeneity in income effects relates to differences in the general physicality of claimants' pre-injury occupation—a proxy for their disutility of work.

Put together, these findings are consistent with a rapidly increasing disutility of work in a population of American workers with permanent partial disabilities. Furthermore, these findings are consistent with a hypothesis that income effects reflect differences in underlying disutility of work across populations. The fact that income effects increase with the speed at which the disutility of work increases has important implications for policy and the interpretation of labor supply responses to disability programs estimated in prior literature. In particular, it is consistent with a significant portion of the labor supply reductions following receipt of disability benefits being due to workers making optimal choices that enhance their utility rather than distortionary disincentives.

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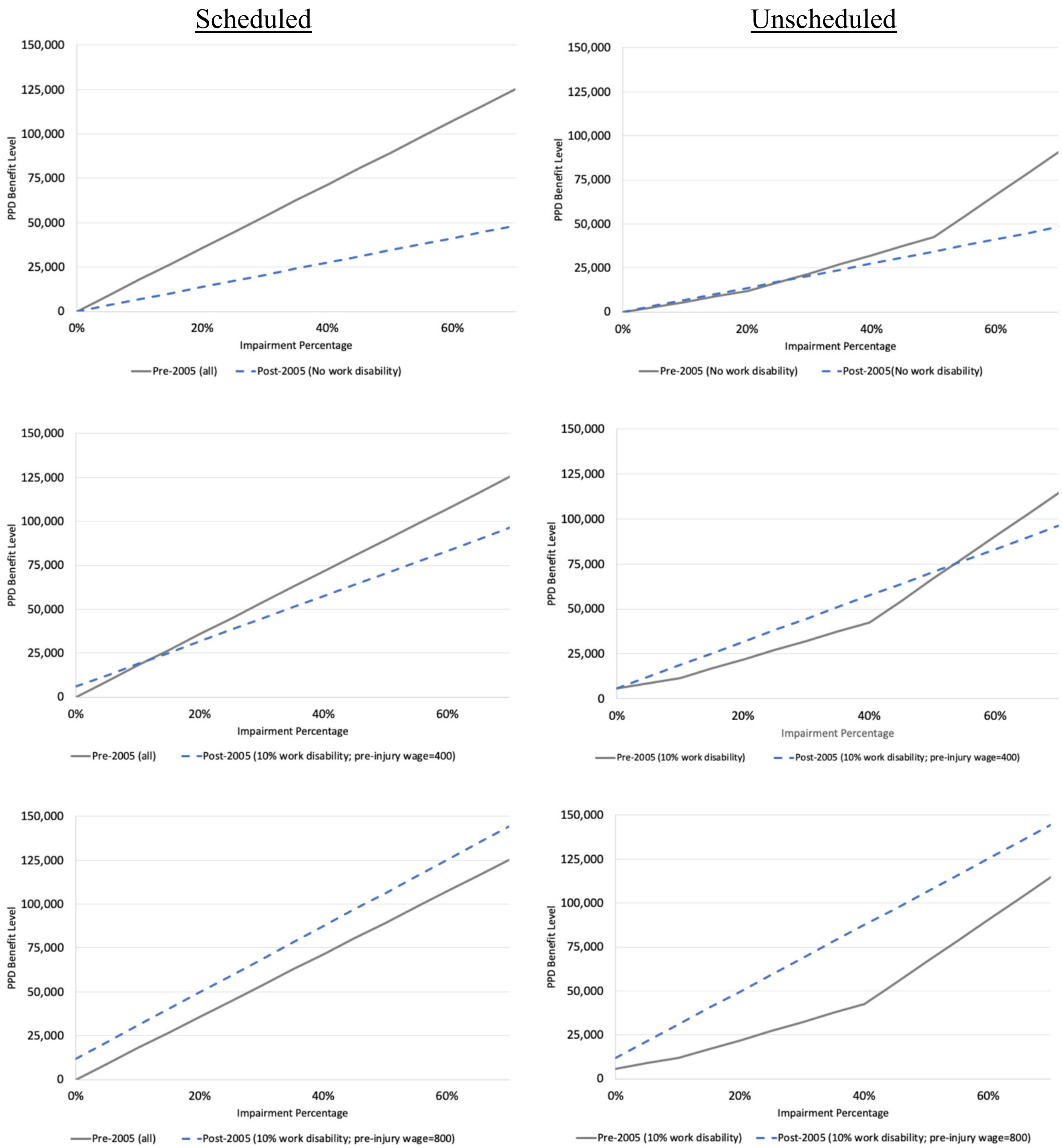
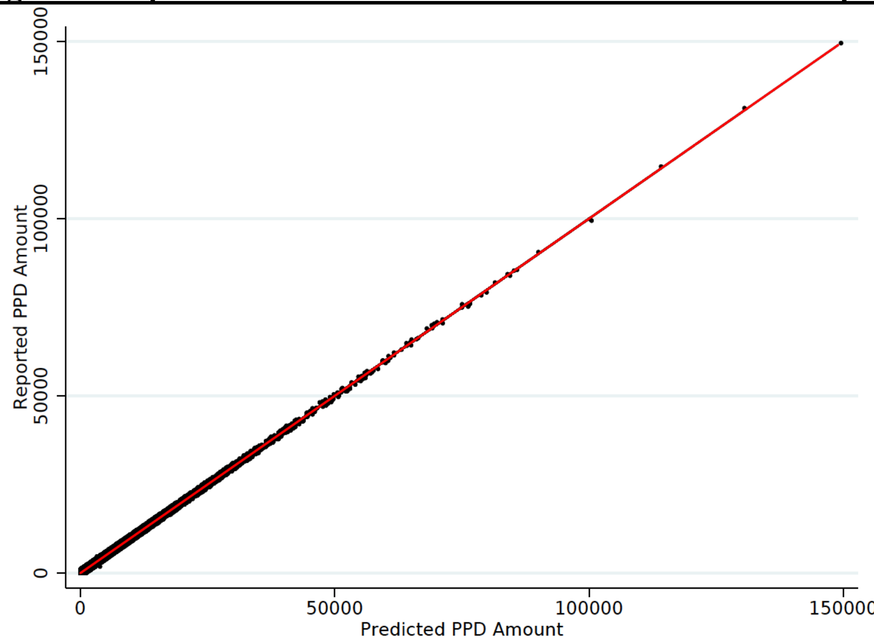
Figure 1: PPD Benefit Levels by Impairment Rating, Before and After Reform

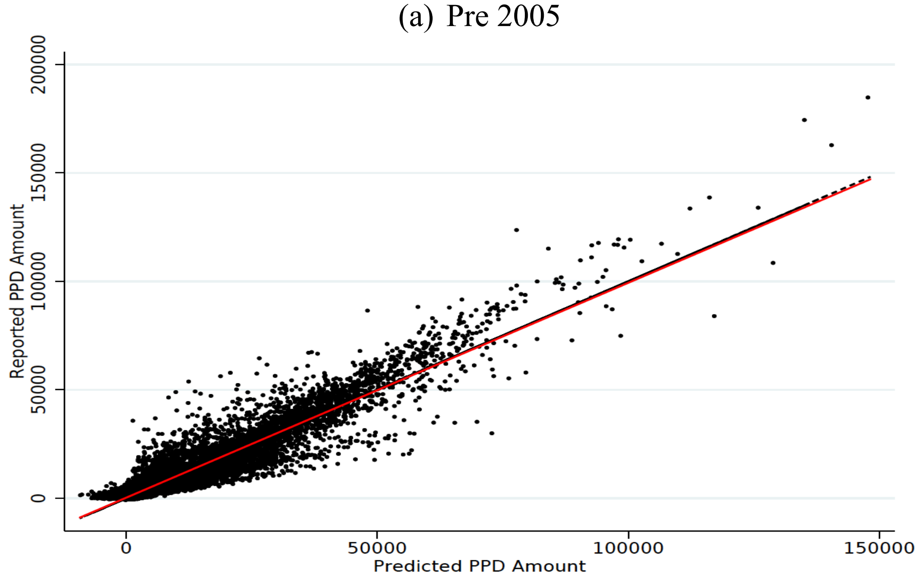


Notes: Author calculations based on Oregon PPD benefit formulas as described in the Oregon Disability Rating Standards, 2015. To obtain these estimates, we calculated the benefit under the old (pre-2005) and new (post-2005) benefit formulas at each possible impairment percentage. Because the formulas prior to 2005 varied depending on whether the injury was classified as scheduled or unscheduled, we calculate the benefits separately using the pre-2005 scheduled formula (left column) and pre-2005 unscheduled formula. (right column). Each subplot shows a benefit calculation using the pre-2005 formula (gray line) and a post-2005 formula (blue dashed line) assuming a specific set of inputs specified in the figure. Each row shows a separate set of inputs related to work disability: no work disability (top row); 10% work disability and pre-injury weekly wage of $400 (middle row); and 10% work disability and pre-injury weekly wage of $800 (bottom row). Prior to 2005, the formula for scheduled injuries only depended on the impairment percentage so the gray line remains the same in each plot in the left column.

Figure 2: Reported vs. Predicted PPD Benefit from Formula Inputs



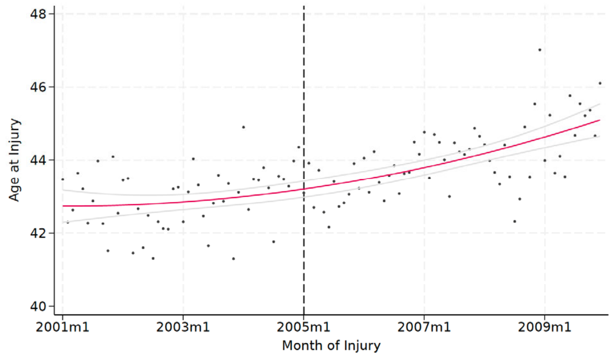
(a) Pre 2005



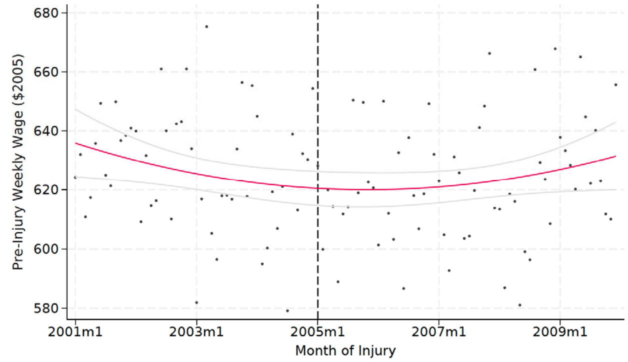
(b) Post 2005

Notes: Based on data from the Oregon Department of Business and Consumer Services, 2001-2009. Figures plot actual benefits as reported in the administrative data (y-axis) against our regression predicting benefits based only on formula inputs that are consistently observed in the data for claims occurring before and after the policy change. These inputs include impairment percentage (interacted with an indicator for scheduled injuries); the sum of impairment and work disability percentages (interacted with an indicator for unscheduled injuries); and a set of observable characteristics that are used to predict work disability including age, gender, medical expenditures, duration of temporary disability benefits, occupation and injury characteristics including nature and body part of injury. All of these inputs are subsequently interacted with year indicators due to the fact that the inputs are used differently in calculating benefits before and after 2005. Figure shows the regression perfectly predicts benefits prior to 2005 because we perfectly observe all formula inputs for the pre-2005 benefit formulas for all claims; there is some random noise in the post-2005 prediction because we do not observe work disability percentages consistently for all claims and therefore include the prediction, rather than actual percentage.

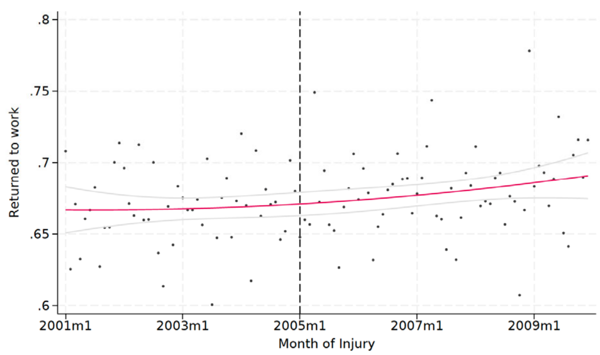
Figure 3: Trends in Formula Inputs by Month of Injury



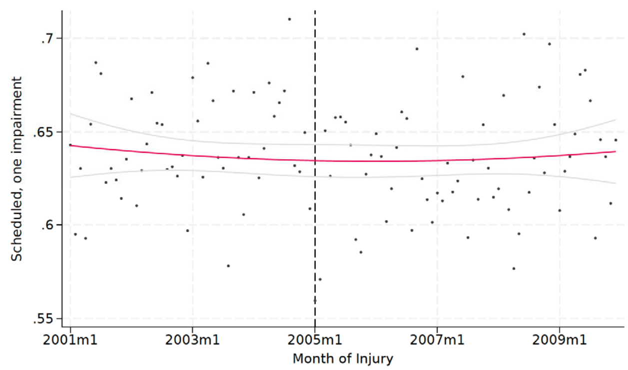
Age at injury



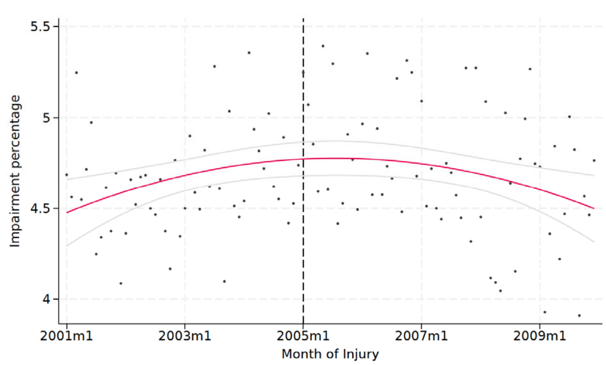
Pre-injury weekly wage



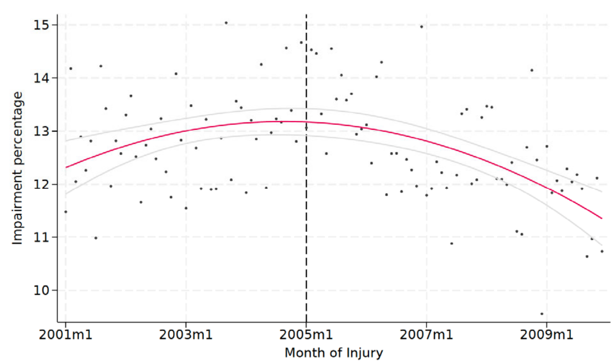
Share returned to work at claim closure



Share of injuries to scheduled body parts



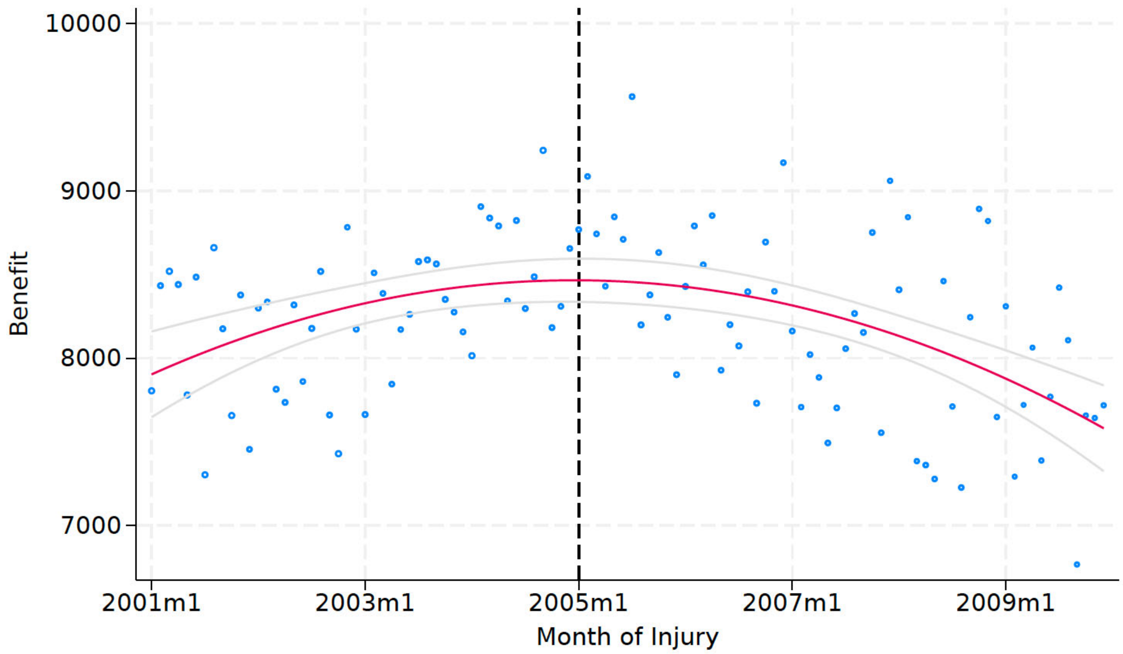
Impairment rating (scheduled)



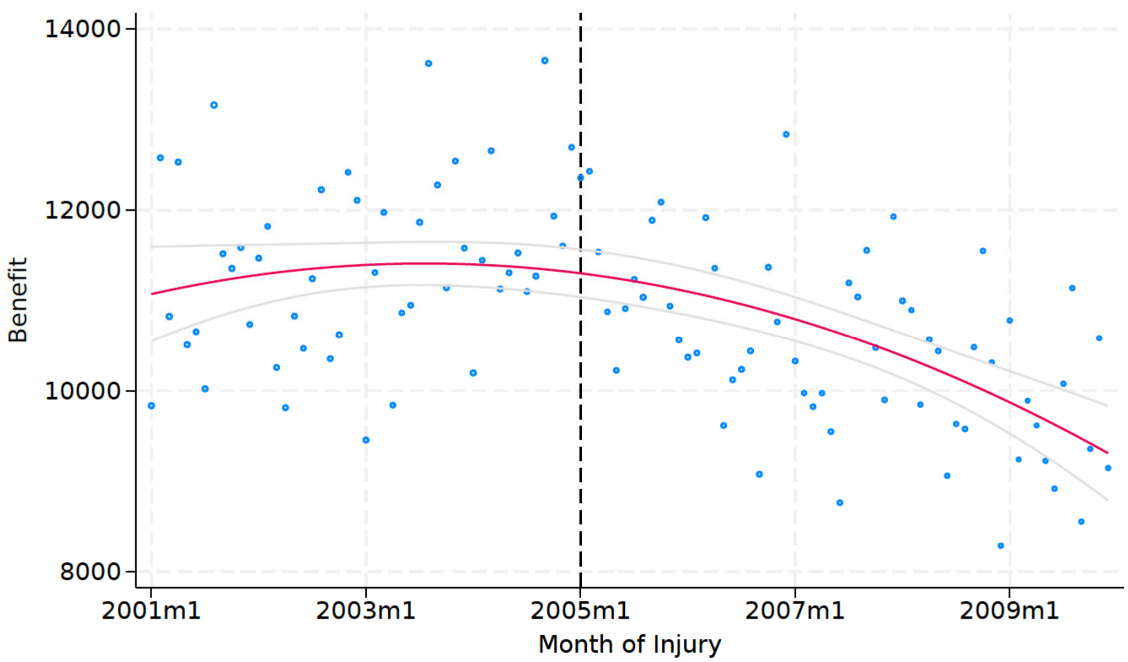
Work disability + impairment rating (unscheduled)

Notes: Data from the Oregon Department of Business and Consumer Services, 2001-2009. Each figure shows a bin-scatter of the value of each demographic characteristic shown in the title, averaging within each month of injury. Colored lines plot the trend and gray lines show 95 percent confidence intervals.

Figure 4: Trend in hypothetical predicted benefits



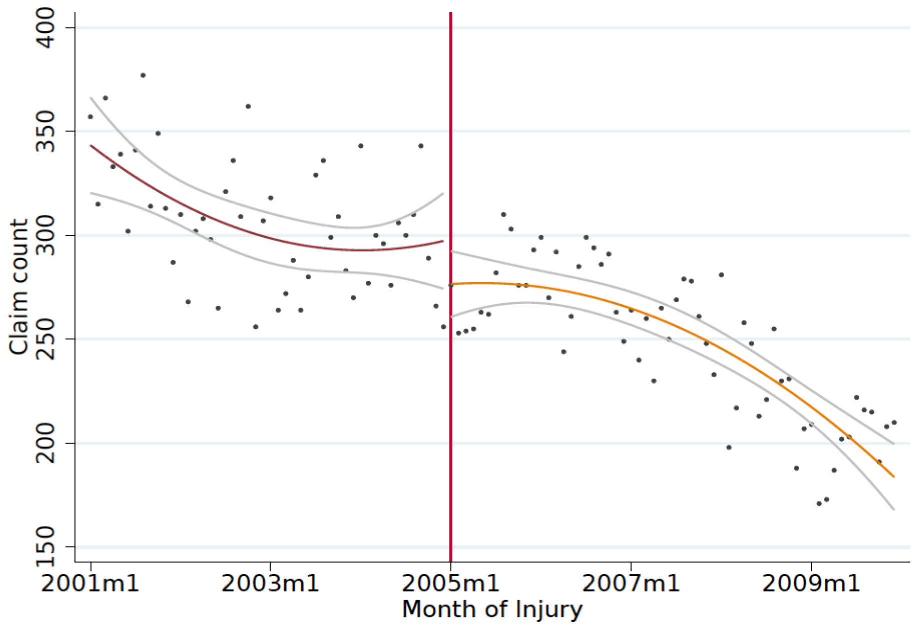
(a) Expected 2004 benefit



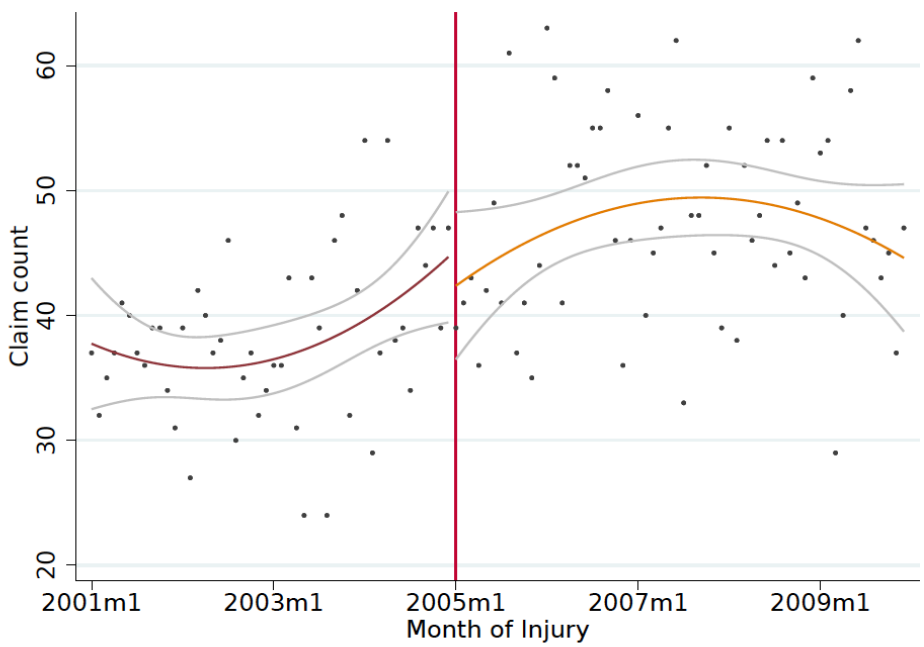
(b) Expected 2005 benefit

Notes: Data from the Oregon Department of Consumer and Business Services, 2001-2009. Figure predicts the value of a hypothetical benefit for all claimants using observed characteristics and ratings, but holding the formula fixed under the pre-2005 formula -using the values in the 2004 formula – and under post-2005 formula – holding constant the values in the 2007 formula. Circles indicate average expected benefit in each month of injury and colored lines plot the trend (with 95-percent confidence intervals in gray)

Figure 5: Claim Frequency by Expected Change in Benefit



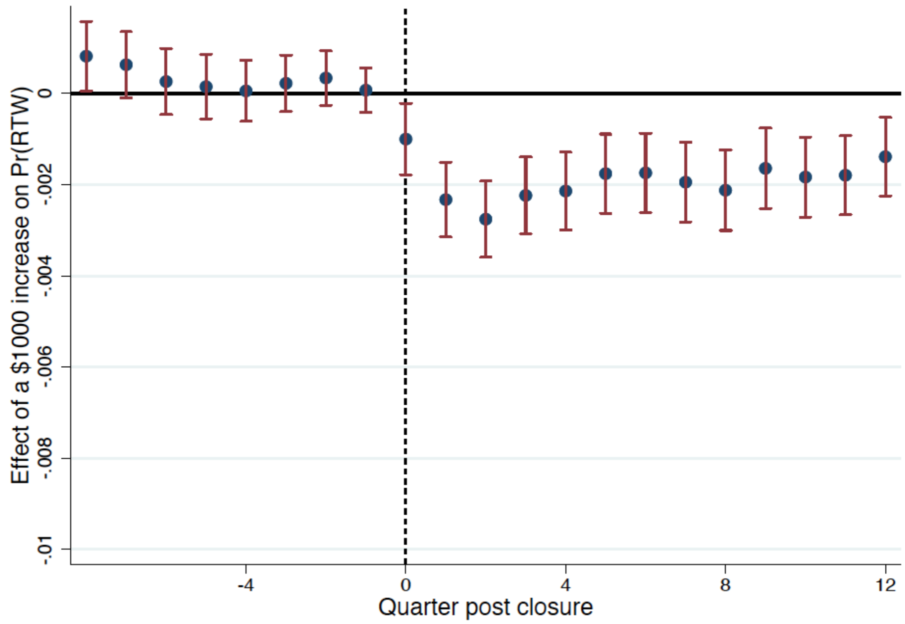
(a) Claims with a greater expected benefit in 2005



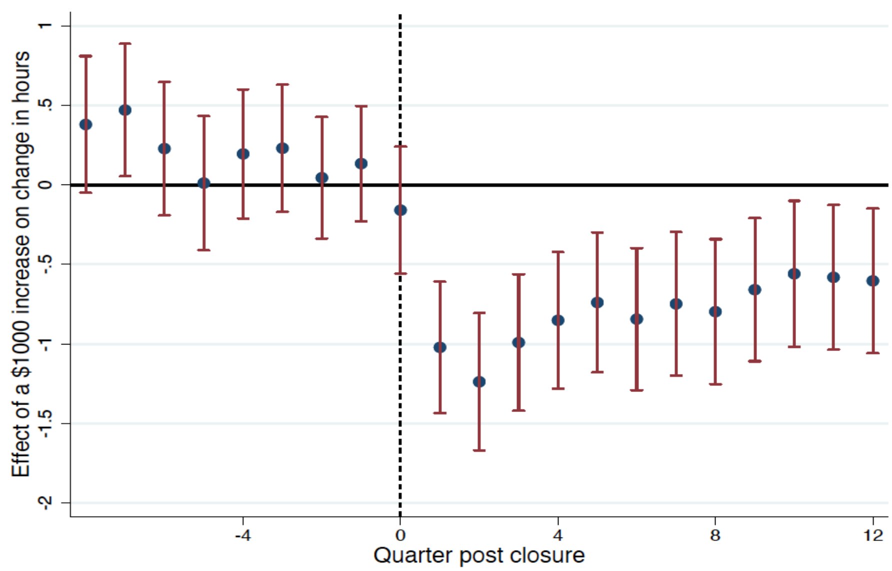
(b) Claims with a smaller expected benefit in 2005

Notes: Based on data from the Oregon Department of Business and Consumer Services, 2001-2009. Each plot shows the count of claims in each month that would be expected to win or lose after the reform based on an estimate of hypothetical benefits. We calculate hypothetical expected benefits for each claim using actual formula inputs in the administrative data, but assume all claims occurred in 2004 and 2005. We then calculate the difference between these two hypothetical benefits to determine which claims would receive larger benefits after the reform compared to their hypothetical benefit prior to the reform, and vice versa. Claims where the expected benefit is larger in 2005 are plotted in (a); claims where the 2005 expected benefit is smaller are plotted in (b). Colored lines plot the trend and gray lines show 95 percent confidence intervals.

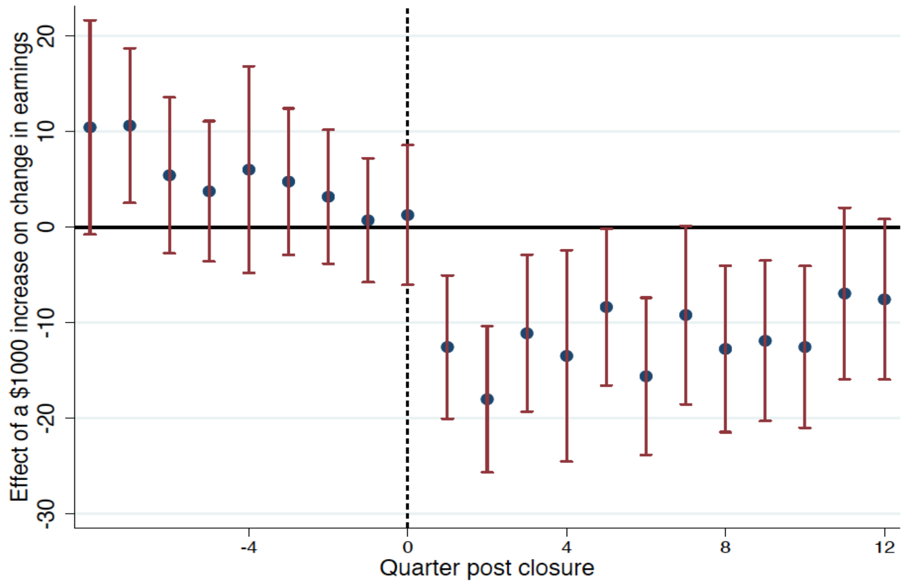
Figure 6: Trends in Estimated Effects over Time



(a) Pr(Return to Work)



(b) Hours



(c) Earnings

Notes: Based on data from the Oregon Department of Business and Consumer Services and Oregon Employment Department, 2001-2010. Each point on the graph is the coefficient from a separate regression from Equation 4 regressing the outcome of interest listed in the figure header for a different quarter before or after claim closure on predicted benefits and formula inputs including impairment percentage (interacted with an indicator for scheduled injuries); the sum of impairment and work disability percentages (interacted with an indicator for unscheduled injuries); and a set of observable characteristics that are used to predict work disability including age, gender, medical expenditures, duration of temporary disability benefits, occupation and injury characteristics including nature and body part of injury. 95 percent confidence intervals shown in the red bars.

Table 1: Effect of SB 757 on Benefits, Select Example Cases

Case	Injury	Impairment Percentage	Impairment + Work Disability Percentages	2004 Benefit	2005 Benefit	Difference
1	Arm (S)	12%	12%	$21,466	$8,263	-$13,203
2	Back (U)	12%	12%	$7,066	$8,263	$1,197
3	Arm (S)	12%	18%	$21,466	$35,263	$13,797
4	Back (U)	12%	18%	$17,664	$35,263	$17,599
5	Back (U)	6%	18%	$14,131	$25,731	$11,600

Notes: Table shows what the PPD benefit would be before and after the reform for five example injuries. Benefits calculated by applying the formulas explained in the text from the Disability Rating Standards-Oregon Administrative Rules outlined in Bulletin 111 by the Oregon Department of Consumer Business Services Workers' Compensation Division. In this table, the impairment percentage reflects the percentage relative to the whole person which, for scheduled injuries, was derived by dividing the specified degrees for the body part(s) by 320 (the maximum number of degrees for the whole body). Weekly wage was assumed to be $600 when calculating the post-2005 benefit amounts.

Table 2: Claim Demographic Characteristics Pre- and Post- Reform

	(1) All	(2) Pre 2005	(3) Post 2005	(4) Difference	(5) P-value
Claimant Characteristics					
Age > 40	0.63	0.62	0.64	0.02	0.00
% male	0.72	0.73	0.71	-0.01	0.00
Pre-injury weekly wage ($2005)	625	627	622	-5	0.09
Medical expenditures ($2005)	12,677	12,573	12,777	204	0.19
TTD days	50	51	49	-2	0.02
% released to work at claim closure	0.81	0.79	0.83	0.04	0.00
% returned to work at claim closure	0.68	0.67	0.68	0.01	0.00
Pre-injury occupation					
Transportation	0.18	0.18	0.18	-0.01	0.20
Production	0.17	0.20	0.15	-0.06	0.00
Construction	0.12	0.12	0.13	0.01	0.01
Maintenance	0.07	0.07	0.08	0.01	0.00
Other Occupation	0.45	0.43	0.47	0.04	0.00
Injury Characteristics					
Fracture/break	0.34	0.34	0.34	0.00	0.49
Strain/sprain	0.33	0.39	0.28	-0.11	0.00
Trauma/unexpected	0.15	0.11	0.19	0.08	0.00
Wounds, cuts, burns	0.08	0.10	0.07	-0.02	0.00
Other	0.09	0.07	0.12	0.05	0.00
Observations	33,778	16,520	17,258		-

Notes: Data from Oregon Department of Consumer and Business Services, 2001-2009. P-values test the equivalence of the means among claims occurring before and after 2005.

Table 3: Formula Inputs and Body Codes, Pre and Post 2005 Reform

	Scheduled Injuries							
	Percentage of All Injuries				Impairment Percentage			
	Pre 2005	Post 2005	Difference	P-value	Pre 2005	Post 2005	Difference	P-value
Overall	62.0	63.6	1.7	0.00	4.6	4.7	0.1	0.38
Arm/Shoulder	30.2	31.1	0.9	0.07	9.2	9.2	-0.1	0.66
Leg/Hip	25.8	25.0	-0.8	0.10	4.1	4.4	0.2	0.01
Hand/Finger	19.2	19.2	0.1	0.88	4.1	4.2	0.1	0.25
Toes/Foot	7.7	6.8	-0.9	0.00	4.3	4.3	0.0	0.78
Ear	0.5	0.5	0.0	0.96	12.7	12.5	-0.2	0.92
Eye	0.1	0.1	0.0	0.73	7.3	8.1	0.8	0.80
	Unscheduled Injuries							
	Percentage of All Injuries				Impairment + Work Disability Percentage			
	Pre 2005	Post 2005	Difference	P-value	Pre 2005	Post 2005	Difference	P-value
Overall	38.0	36.4	-1.7	0.00	12.9	12.5	-0.3	0.09
Low back	13.9	11.1	-2.8	0.00	14.5	14.4	-0.1	0.85
Neck	4.3	3.4	-0.9	0.00	13.1	13.4	0.3	0.58
Back - multiple	2.1	2.0	-0.1	0.51	11.1	10.6	-0.6	0.47
Other body systems	0.8	0.6	-0.1	0.10	12.9	13.5	0.6	0.70
Brain	0.3	0.6	0.2	0.00	21.8	20.7	-1.1	0.72

Notes: Data from Oregon Department of Consumer and Business Services, 2001-2009. P-values test the equivalence of means among claims before and after 2005. Some arm, shoulder, leg and hip injuries are classified as scheduled and unscheduled prior to 2005, we combine them and list under the scheduled category for simplicity in this table. Total percent of all injuries summed across categories exceeds 100 percent prior to 2005 due to claims with multiple injuries.

Table 4: IV Estimates of the Effect of PPD Benefit on Labor Supply in Third Year Post-Closure: Main Results

	(1) Return to Work	(2) Hours	(3) Earnings
PPD Benefit/1000	-0.00188***	-2.36***	-38.56**
	(0.0004)	(0.86)	(16.20)
Predicted Y-mean at average benefit	0.726	1127	20714
Observations	33,778	33,778	33,778
R-squared	0.10	0.15	0.36
First stage F-statistic	233.5	233.5	233.5

Notes: Data from Oregon Department of Consumer and Business Services and Oregon Employment Department, 2001-2009. Table shows IV coefficients on PPD benefits from the second stage of the specification described in Equation 4. Additional controls in the second stage regression include the following: for scheduled injuries, impairment rating and case characteristics, respectively, interacted with pre-injury wage; and for unscheduled injuries, the sum of impairment and work disability rating interacted with pre-injury wage, and uninteracted case characteristics. For case characteristics, we include interactions between variables that are strong predictors of work disability eligibility and those that are strong predictors of work disability ratings. Robust standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 5: Summary of Prior Studies on Effect of a One-Time Increase in Non-Labor Income on Labor Supply

Study	Population	Effect of $1,000 on Employment
Cesarini et al. (2017)	Swedish lottery players	-0.014
Marie and Vall Castello (2012)	Spanish DI PPD beneficiaries at age 55	-0.023✢
Giupponi (2019)	Italian survivors' insurance beneficiaries	-0.034
Jacob and Ludwig (2012)	Chicago housing voucher recipients	-0.035✢
Golosov et al. (2021)	U.S. lottery players	-0.037
Vivalt et al. (2024)	U.S. individuals with low incomes	-0.061✢
Gelber, Moore and Strand (2017)	U.S. SSDI beneficiaries	-0.099✢
Deuchert and Eugster (2019)	Swiss DI PPD beneficiaries at age 50 with disability degree between 67 and 69%	-0.132✢✢
Autor et al. (2016)	U.S. VA DC beneficiaries with type 2 diabetes	-0.159✢
Mullen and Rennane (2025) (this study)	Oregon WC PPD beneficiaries	-0.188

Notes: DC=disability compensation, DI=disability insurance, PPD=permanent partial disability, WC=workers' compensation. Income effects are scaled to represent the percentage point change in employment resulting from a one-time $1,000 change in non-labor income.

✢To convert increase in annual DI benefit/cash value of housing voucher to a one-time payment, we assume a discount rate of 2.4%.

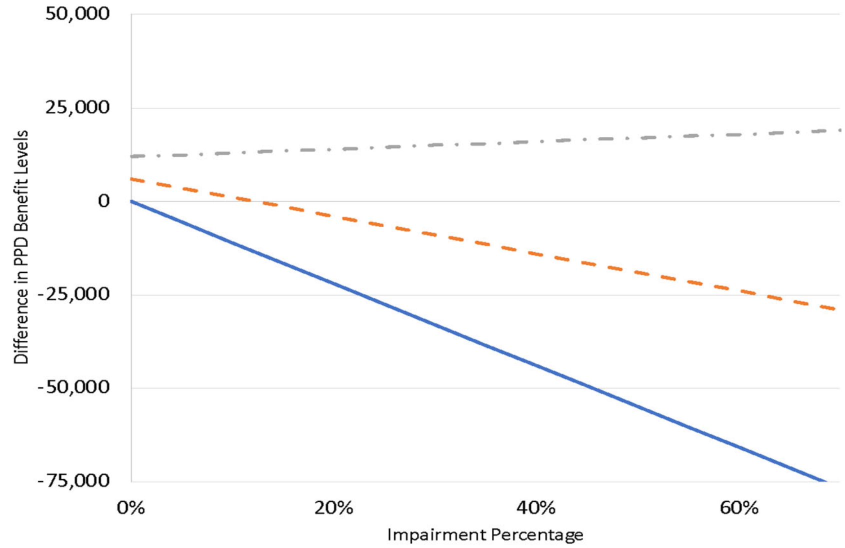
✢✢Midpoint between lower and upper bound estimates (-0.081 and -0.182).

Table 6: IV Estimates of the Effect of PPD Benefit on Labor Supply in Third Year Post-Closure: By Physicality of Occupation

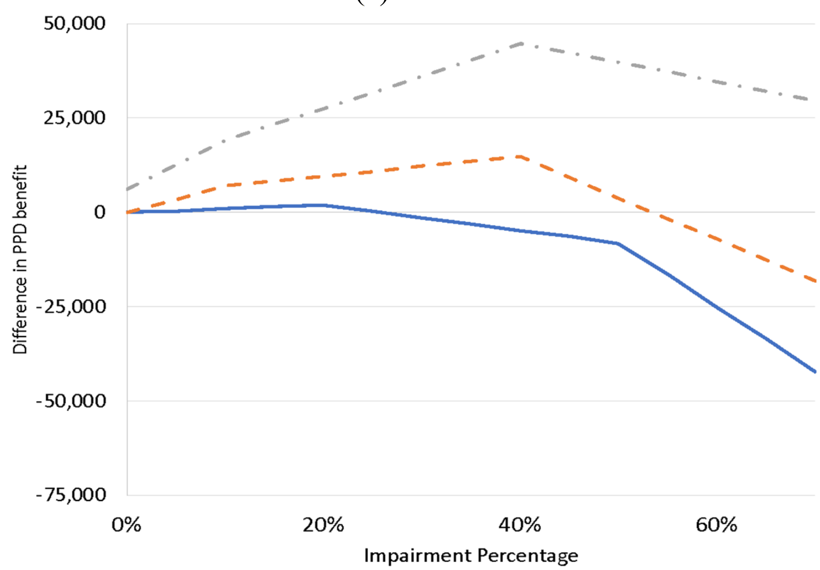
	Return to Work	Hours	Earnings	Quartile N
Q1 (low physicality)	-0.00110	-0.81	22.33	8,493
	(0.00099)	(1.86)	(38.83)	
Q2	-0.00202***	-2.77**	-53.52**	13,194
	(0.00071)	(1.41)	(22.87)	
Q3	-0.00244*	-3.24	-41.97	3,154
	(0.00143)	(2.78)	(44.63)	
Q4 (high physicality)	-0.00228**	-3.29*	-76.05*	7,527
	(0.00099)	(1.88)	(41.86)	
Observations	31,933	31,933	31,933	
P-value of equivalence of all quartiles	0.808	0.774	0.301	
P-value of equivalence of Q1 vs Q4	0.397	0.347	0.085	
Y-mean Q1	0.751	1,135	22,049	
Y mean Q2	0.725	1,156	19,242	
Y mean Q3	0.687	1,040	14,842	
Y mean Q4	0.727	1,128	25,000	

Notes: Data from Oregon Department of Consumer and Business Services and Oregon Employment Department, 2001-2009. Table shows IV coefficients on PPD benefits from the second stage of the specification described in Equation 4 run on a regression interacting the PPD benefit with a measure of the physicality of the occupation. We use the O*Net level score for Performing General Physical Activities of various occupations, binned into quartiles. Because we measure occupations at the 2-digit level, the distribution of quartiles results in uneven sized bins, indicated by the "Quartile N" column. Outcomes are listed in headers. Additional controls in the second stage regression include the following: for scheduled injuries, impairment rating and case characteristics, respectively, interacted with pre-injury wage; and for unscheduled injuries, the sum of impairment and work disability rating interacted with pre-injury wage, uninteracted case characteristics, and injury year FE. For case characteristics, we include interactions between variables that are strong predictors of work disability eligibility and those that are strong predictors of work disability ratings. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Figure A1: Difference in PPD Benefit Levels by Impairment Rating (Percent of Person), Before and After Reform



(a) Scheduled

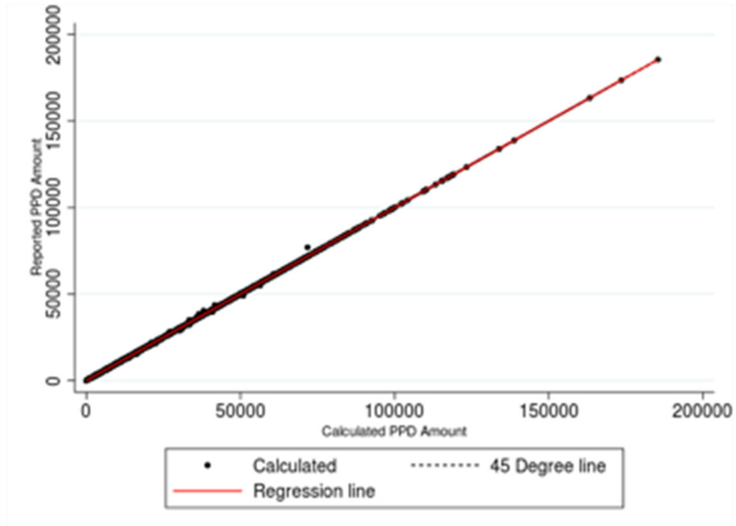


(b) Unscheduled

No work disability — 10% work disability, pre-injury wage=400 — 10% work disability, pre-injury wage=800

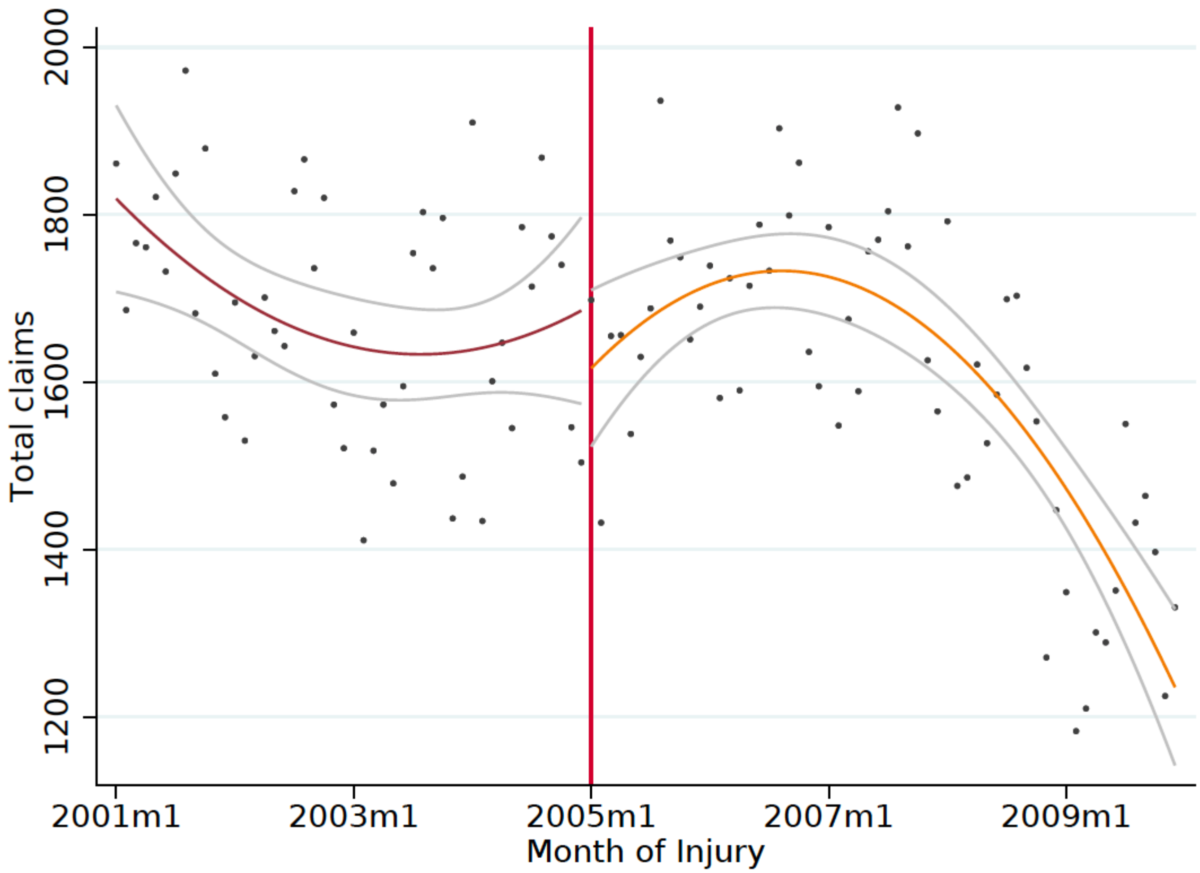
Notes: Author calculations based on Oregon PPD benefit formulas as described in the Oregon Disability Rating Standards, 2015. To obtain these estimates, we calculated the benefit under the old (pre-2005) and new (post-2005) benefit formulas at each possible impairment percentage. These benefit values are shown in Figure A1. We subtracted the new value from the old value to plot the difference shown above. (In practice we do not observe impairment percentages above 60% so have truncated the figure to a relevant range) In addition to variation in the impairment percentage (shown along the x-axis), benefits depend on the work disability percentage in both regimes and also depend the workers' pre-injury wage in the post-2005 regime. We therefore repeat the calculation for three hypothetical scenarios which vary these parameters: a worker with no work disability (solid blue line); a worker with 10% work disability and pre-injury weekly wage of $400 (dashed orange line); and a worker with 10% work disability and pre-injury weekly wage of $800 (dashed gray line). Because the formulas prior to 2005 varied depending on whether the injury was classified as scheduled or unscheduled, we plot the hypothetical difference separately using the pre-2005 scheduled formula in (a); and the pre-2005 unscheduled formula in (b).

Figure A2: Calculated vs. Reported Actual Benefit (post-2005 claims)



Notes: Based on data from the Oregon Department of Business and Consumer Services, 2001-2009. Figures plot actual benefits as reported in the administrative data (y-axis) against our regression predicting benefits only for the post-2005 claims. Because we are only using post-2005 claims in this plot, we include all actual formula inputs, including those not observed in claims prior to 2005 (and thus not included in our main specification). This figure demonstrates that our model perfectly predicts workers' benefits when using all actual inputs.

Figure A3: Trends in Overall Claim Volume by Month of Injury



Notes: Data from the Oregon Department of Business and Consumer Services, 2001-2009. Figure shows a total count of each month of injury. Colored lines plot the trend and gray lines show 95 percent confidence intervals.

Appendix Table 1: Predictors of Work Disability Eligibility

	(1)	(2)
Return to work at closure	-0.06***	-0.05***
	(0.00)	(0.00)
Released to work at closure	-0.75***	-0.73***
	(0.00)	(0.00432)
TTD Duration Deciles (omit lowest decile)		
2.decile		-0.00
		(0.01)
3.decile		-0.00
		(0.01)
4.decile		0.00
		(0.01)
5.decile		-0.00
		(0.01)
6.decile		0.00
		(0.01)
7.decile		-0.00
		(0.01)
8.decile		0.00
		(0.01)
9.decile		0.02***
		(0.01)
10.decile		0.08***
		(0.01)
Constant	0.80***	0.76***
	(0.00)	(0.01)
Observations	17,708	17,708
R-squared	0.76	0.77
Ymean	0.139	0.139

Notes: Data from Oregon Department of Consumer and Business Services,

Appendix Table 2: Predictors of Work Disability Percentage

	(1)	(2)
Age > 40	2.32***	2.35***
	(0.27)	(0.27)
Male	-0.62*	-0.63*
	(0.34)	(0.34)
Traumas	-0.10	-0.10
	(0.43)	(0.43)
Fractures/breaks	-0.40	-0.40
	(0.38)	(0.38)
Sprain/Strain	-0.40	-0.39
	(0.35)	(0.35)
Wound/cut/burn	0.33	0.39
	(0.87)	(0.87)
Medical Expenditures (1000s, $2012)	0.02***	0.02***
	(0.01)	(0.01)
Body System - First Impairment		
Upper extremities	4.76**	4.69**
	(2.23)	(2.23)
Shoulder	4.30*	4.20*
	(2.23)	(2.23)
Lower extremities	2.82	2.71
	(2.23)	(2.23)
Back	4.57**	4.47**
	(2.23)	(2.23)
Head	3.37	3.28
	(2.27)	(2.27)
Internal	1.82	1.67
	(2.56)	(2.56)
Body System - Second Impairment		
Upper extremities	0.50	0.27
	(1.68)	(1.68)
Shoulder	0.65	0.14
	(3.20)	(3.20)
Lower extremities	-2.79	-2.88
	(2.20)	(2.21)
Back	-1.46	-1.72
	(2.78)	(2.78)
Head	1.24	1.08
	(3.12)	(3.12)

Appendix Table 2, con't: Predictors of Work Disability Percentage

Occupation Categories		
o1	-2.28**	-2.37***
	(0.90)	(0.90)
o2	1.10	1.44
	(2.88)	(2.89)
o3	-5.72	-4.57
	(6.22)	(6.25)
o4	-5.24***	-5.49***
	(1.72)	(1.72)
o5	-2.68	-2.47
	(2.12)	(2.12)
o6	-3.19	-3.38
	(2.57)	(2.59)
o8	-3.37**	-3.29**
	(1.36)	(1.37)
o9	-3.73*	-3.72*
	(2.11)	(2.11)
o10	-4.63***	-4.74***
	(0.88)	(0.88)
o11	0.11	0.01
	(0.78)	(0.78)
o12	-1.98	-2.06
	(1.28)	(1.28)
o13	-0.43	-0.35
	(0.82)	(0.82)
o14	1.39**	1.30*
	(0.70)	(0.70)
o15	-3.28**	-3.49**
	(1.54)	(1.54)
o16	-1.04	-0.93
	(0.72)	(0.72)
o17	-2.24***	-2.26***
	(0.82)	(0.82)
o18	4.88***	4.88***
	(0.77)	(0.77)
o19	-2.10***	-2.19***
	(0.55)	(0.55)
o20	-2.91***	-2.88***
	(0.64)	(0.64)
o21	-0.26	-0.37
	(0.56)	(0.56)
o22	0.87*	0.79
	(0.52)	(0.52)

Appendix Table 2, con't: Predictors of Work Disability Percentage

	TTD Duration (omit lowest decile)	
TTD Duration/10	0.05***	
	(0.01)	
2.decile		-1.50
		(1.13)
3.decile		0.01
		(1.14)
4.decile		-1.22
		(1.07)
5.decile		-0.36
		(1.06)
6.decile		1.07
		(1.01)
7.decile		-0.86
		(1.00)
8.decile		0.54
		(0.95)
9.decile		0.49
		(0.91)
10.decile		1.25
		(0.90)
Constant	5.74**	6.24***
	(2.26)	(2.41)
Observations	2,445	2,445
R-squared	0.15	0.15
Ymean	11.44	11.44

Notes: Data from Oregon Department of Consumer and Business Services,

Appendix Table 3: IV Estimates of the Effect of PPD Benefit on Labor Supply in the Third Year Post-Closure: Log Specification

	(1) Log Hours	(2) Log Earnings
PPD Benefit/1000	-0.0001	-0.0010
	(0.0012)	(0.0013)
Predicted Y-mean at average benefit	7.1	9.9
Pct change in Y-mean	-0.01%	-0.10%
$1000 change as pct of average benefit	11.2%	11.2%
Elasticity	-0.001	-0.009
Observations	24,316	24,306
R-squared	0.1013	0.2506
First stage F-statistic	318.3	475.9

Notes: Data from Oregon Department of Consumer and Business Services and Oregon Employment Department, 2001-2009. Table shows IV coefficients on PPD benefits from the second stage of the specification described in Equation 4. Additional controls in the second stage regression include the following: for scheduled injuries, impairment rating and case characteristics, respectively, interacted with pre-injury wage; and for unscheduled injuries, the sum of impairment and work disability rating interacted with pre-injury wage, and uninteracted case characteristics. For case characteristics, we include interactions between variables that are strong predictors of work disability eligibility and those that are strong predictors of work disability ratings. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1